Substituting Reference Solvent σ-Profiles for Learned Ones Degrades COSMO-SAC Solubility Prediction

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Abstract

Physics-informed models route predictions through physical quantities. Accurate references should help. We measure that for solid–liquid solubility: a graph network predicts the charge-density profiles a database tabulates. COSMO-SAC's fixed residual term and a crystal term turn two into a solubility. Supervising the profile against that database moves $\ln x_2$ mean absolute error from 2.037 to 1.927 over five leak-free seeds, a mean without a sign: four gain, one loses. Substituting the reference for the learned one raises the error to 2.336 at inference at every seed, the training-time rise smaller, at one co-adapted seed. Only the substitution has a direction, and neither size can be ordered. On infinite-dilution activity data another database bounds a deployed thermodynamic model's error from below, its inputs' from above. The input bound barely moves (an estimator property, not chemistry) while the error does. One admissibility rule leaves glycol-ether solvents standing, 182 rows over 4 solvents and 19 hydrocarbon and halobenzene solutes, its error exceeding its inputs' by 2.04 (90% interval $[+1.25, +2.87]$), and 1.76 under the pre-declared estimator. Both are lower bounds. Every other subset falls, the set whole with them, and solubility resists the split. That set is a 59-fold search's survivor rather than an established effect: chemistry-blind relabelling reaches its margin in a tenth of draws under the first bound, a seventh under the other. The profile departs from its reference along a shared direction, on separate three-seed runs. A Hammett pK_a closure reverses the substitution's sign in distribution, where its constants suffice.

1 Introduction

How much of a solid dissolves in a given liquid decides whether a drug can be formulated as a tablet or an injection, which solvent a crystallisation is run in, and how a product is purified. Each value has to be measured one solute, one solvent and one temperature at a time. Most industrially relevant combinations therefore remain unmeasured, and independent determinations of the same system disagree by enough to floor what any predictor can achieve. Dissolving a solid also trades the cost of breaking up a crystal against the cost of mixing the freed molecules into the solvent. A predictor has to be right about two unrelated kinds of chemistry at once, on molecules unlike any it was shown.

Predictors divide by how much physical structure they impose. Direct models regress solubility from descriptors or from a representation learned off the molecular graph. FastSolv[1] does so with a deep neural network and leads on extrapolation to new solutes. Such models are the more accurate there, and they are opaque: the prediction arrives as a single number, silent about which part of the chemistry it captured. Physics-informed models instead predict the thermodynamic quantities the physical picture names (an activity coefficient for the solute–solvent interaction, a melting temperature and heat of fusion for the crystal) and combine them through a fixed equation. SolProp[2] does that with a thermodynamic cycle of separately learned components. What that buys is decomposability: each part is a quantity chemists tabulate on its own, so it can be supervised against reference data that a direct regressor is unable to use. On accuracy the black box leads and the physics-informed cycle trails it. Learned surrogates for the activity coefficient have advanced quickly, and they split the same way. Winter et al.[3] predict $\ln \gamma_2^\infty$ from SMILES with a language model, and Sanchez-Medina et al.[4] and Wang et al.[5] embed a Gibbs–Helmholtz constraint in a graph network for its temperature dependence. All three make the activity coefficient the regression target. A second line keeps a mechanistic model and learns its input instead. COSMO-SAC[6] is a segment-activity reformulation of Klamt's COSMO-RS. The σ-profile it consumes (the screening-charge density over a molecule's cavity) is that framework's construct, tabulated from quantum chemistry.[7] Predicting it from structure removes that calculation from the pipeline. TeNNet-SAC[8] does so while also learning the segment-activity kernel and the cavity geometry, and reports accuracy better than COSMO-SAC's after end-to-end fine-tuning. That is the opposite headline from this one, on the same broad construction, and we set the two side by side in §3.3.

When a fixed physical structure improves or degrades a learned model is the broader question of physics-informed machine learning.[9] Its dominant forms impose structure as a differentiable object: a governing equation enforced through the loss,[10] gray-box hybrids composing learned components with fixed mechanistic maps,[11] and operator learning with a trainable correction to a misspecified equation.[12] The prevailing expectation is that more physics improves generalisation, yet the same structure has documented failure modes in which it makes optimisation or accuracy worse.[13] In neural PDE solvers, a learnable term added to a fixed discretised operator absorbs truncation artifacts rather than physics, so the apparent interpretability is illusory[14] (though that error source is numerical, which puts it beyond external reference). The same concern appears in *concept-bottleneck models*: networks routed through a layer of named, human-interpretable concepts, so that those concepts are the whole of what the prediction is made from. That is the same shape as a learned intermediate feeding a fixed physical map, and there the failure is called *concept leakage*: a learned concept silently encodes information beyond its nominal meaning.[15–17] Recent work traces one route to it to the downstream model. A misspecified head forces the encoder to deform the concept into carrying a dependence beyond that head's representational reach,[18] and overwriting a frozen head's leaked concepts with ground truth can then degrade accuracy.[19] The gap between a fixed physical map and reality is the model-discrepancy problem of Kennedy and O'Hagan[20] and its calibration critique.[21] Four further literatures (forecast decomposition, quasi-maximum likelihood, identifiability, and the validity of composed pipelines) supply pieces we use below.

The design rests on an assumption rarely stated because it is taken for self-evident: a model supplied with physical inputs closer to their true values predicts better. For one class of intermediate that assumption can be tested outright, because a reference value is tabulated and can be handed to the model in place of the one it computed. It needs testing. When a predictor of this shape is inaccurate, either the fixed physical model is wrong or the learned quantities feeding it are uninformative, and the two call for opposite remedies. What this literature lacks is a way to *measure*, rather than diagnose, which of the two limits a composed predictor's accuracy, and whether the learned intermediate has stopped reporting the quantity it is named after. In the concept-bottleneck case the external reference that would settle either question is missing. We supply one here (a reference σ-profile for solubility, tabulated substituent constants for a second chemical domain) and hold the mechanistic model fixed, so the tabulated value stays available as an external reference for the learned one.

Here a graph neural network predicts the pair of charge-density distributions COSMO-SAC consumes, in place of the quantum-chemical calculation that ordinarily supplies them. The tabulated counterpart can enter at either of two points: as a target the network is trained toward, or as a replacement for what the network computed at the moment of prediction (Fig. 1). Those two entry points are the two operations the word *grounding* names, and here only one of them carries a sign. Training the network toward the reference moves the error four ways out of five and the fifth the other way; supplying the reference at the moment of prediction raises it at every seed. Tabulated profiles serve a third purpose on activity-coefficient data, where they divide the error of a fixed model between the model itself and the inputs handed to it. Solubility withholds that division, because a solubility measurement constrains the crystal and the mixing contributions jointly and leaves the blame between them unassigned. What the division returns is a map rather than a verdict: one answer per stratum, a stratum being a chemically defined set of rows. Whether a fixed model's own error stands above what its inputs fail to carry is a property of the chemistry the model is asked about. Where it does stand above, a more accurate input exposes the model's bias instead of removing it. The learned distribution meanwhile drifts away from its tabulated counterpart along a direction shared across molecules, and stops reporting the quantity it is named after. Tabulated substituent constants behave the same way inside a fixed Hammett relation, the linear equation that predicts how readily an acid gives up its proton from the constants tabulated for its substituents.

This article is organised as follows. §2 sets out the composed predictor, the identity that divides its error between the fixed model and the inputs, and the estimator for each part. It then gives the data, the measurement noise that floors what any predictor can reach, and the evaluation protocol. §3 reports the signs of the two grounding operations, the division of the error on activity-coefficient data and the chemical map it returns, and the drift of the learned distribution away from its reference. It then runs the same two measurements on a second fixed model for acid dissociation, places our accuracy against published models, and states the limitations of all of it. §4 states the design rule our measurements support.

2 Computational methods

2.1 The composed predictor and its error split

For a solute s in solvent v at temperature T, the target is $y = \ln x_2$, the log saturated mole fraction. Dissolving a solid trades breaking the crystal lattice (Φ) against mixing the freed molecules into the solvent ($\ln \gamma_2$), and solid–liquid equilibrium (SLE) sets the log-solubility to their negative sum:

$$\ln x_2 = -\underbrace{\frac{\Delta H_{\mathrm{fus}}}{R}\Big(\tfrac{1}{T} - \tfrac{1}{T_m}\Big)}_{\Phi(T)\ \text{(crystal)}} - \underbrace{\ln \gamma_2(x_2, T)}_{\text{activity}}, \quad (1)$$

with no constant-ΔC_p correction (ΔC_p being the difference in heat capacity between the liquid and the solid). That term is absent from the displayed equation because it is absent from the deployed model, which fixes $\Delta C_p = 0$, the head that would predict it being off in the pinned configuration. The exposure is one-signed and substantial. At the group-contribution value this corpus carries, $60\,\mathrm{J\,mol^{-1}\,K^{-1}}$, the omitted term is a median 0.32 and a mean 0.40 in $\ln x_2$ over the 1318 labelled test rows that carry a measured T_m, 31 solutes, positive on every one of them. That is the size of the substitution penalty of §3.1 and twice the supervision gain. It enters the physics arms and leaves the closure-free control untouched, and it cancels exactly in the evaluation-time substitution, where Φ is common to the two arms (§S1). It

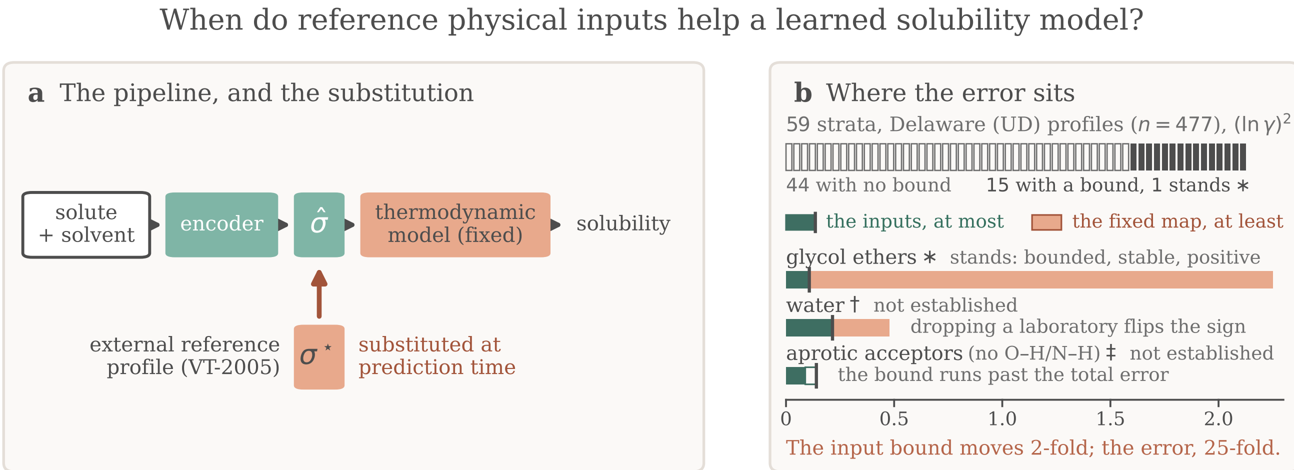

Figure 1: **Overview.** **(a)** A learned encoder predicts a charge-density profile $\hat{\sigma}$ that a fixed thermodynamic model turns into a solubility, and the arrow marks the substitution of the reference $\sigma^\star$ for it. **(b)** The counterpart error on infinite-dilution activity data: the strip is every stratum searched, the bars the three solvent classes with a bound. The starred class alone stands, each mark carrying its own reason, and each block is a one-sided bound rather than an interval. The right-hand block is the error of the map as deployed, the fixed kernel read at a computed profile, and the profile calculation's share of it stays unbounded here. Training-time supervision is omitted.

survives into the training of the σ head, which is fitted through a crystal term carrying a bias of that size and in that direction. §3.3 reads the consequence. The activity term is produced by a *differentiable closure* g. A graph encoder predicts a σ-profile (the distribution of screening charge density over a molecule, the histogram a COSMO model consumes) for each molecule, and COSMO-SAC,[6] a segment-activity reformulation of COSMO-RS, maps the pair of profiles to $\ln \gamma_2$. The segment-interaction energy inside that map (the electrostatic-misfit and hydrogen-bonding terms setting what two surface segments cost each other) is the closure's *kernel*, the part the published revisions rewrite. A closure variant is therefore named for the year of its kernel: the *2002 kernel* this pipeline runs, and the *2010 kernel* that types segments as hydrogen-bond donors or acceptors instead of treating hydrogen bonding with one parameter. Write $z^\star$ for the *reference* physical latent, a tabulated σ-profile, and $\hat{z}$ for the head's own prediction. **Two profile databases supply $z^\star$ below and they are never interchangeable**. *VT-2005*, the Virginia Tech 2005 tabulation,[7] is what the σ head is supervised against and what is substituted for $\hat{z}$ at prediction time. *UD*, the University of Delaware successor tabulation, is what the activity-coefficient rows of §3.5.1 are matched to. Every set, margin and figure below names which of the two it is computed on, and we read no quantity across them. The one place they are deliberately set against each other is the database-to-database bracket of §3.5.3. Feeding $z^\star$ in place of $\hat{z}$ (a σ-oracle) isolates the closure from encoder error: if the closure were well specified, $g(z^\star)$ should be the best available activity estimate. Each trained configuration, and each closure variant evaluated, is an *arm*. A *DirectGNN* control arm drops the closure and emits $\ln x_2$ directly, and we tuned the two separately (§2.4).

We state everything below for a general *composed predictor* $\hat{y} = g(h(x))$ in which a fixed closure g consumes a learned physical intermediate $z = h(x)$, against a matched free predictor that shares the encoder and drops g. Solubility is one instance, and a second is the closure of §3.6 for pK_a, the acid dissociation constant on a negative logarithmic scale.

Solubility alone underdetermines the crystal/activity split of Eq. (1). At infinite dilution with $\Delta C_p = 0$, the observable is affine in $1/T$, and the activity intercept and slope already span that plane, so the profiled Fisher information for ΔH_{fus} is exactly zero and the indeterminacy set is two-dimensional. At finite dilution, the composition factor $(1-x_2)^2$ leaves an information of order x_2^2. The deficiency closes under a rank-completing external constraint and nothing weaker: a direct crystal label, or a fixed activity slope (Lemma S1, §S2). That is why an under-determined factor needs external single-component data and why we expect compensation between the factors. It is silent by itself on whether grounding helps.

2.1.1 The closure–insufficiency decomposition

Let m be the closure's target (here the experimental infinite-dilution activity coefficient (IDAC) $\ln \gamma_2^\infty$, in §3.6 the measured $\mathrm{p}K_a$) and $g(z^\star)$ the closure evaluated on the reference latent.

Lemma 1 (Exact closure–insufficiency decomposition and its one-sided bounds)**.** *Let g be a* fixed *(non-fitted) closure and let $z^\star$ denote its* entire *argument (the profile pair together with the temperature wherever the closure reads one). Then* (a) *the reference-input mean squared error splits exactly,*

$$\underbrace{\mathbb{E}\big[(m-g(z^\star))^2\big]}_{\textit{ref.-input MSE}} = \underbrace{\mathbb{E}\big[\mathrm{Var}(m \mid z^\star)\big]}_{B_{\text{insuff}}\ (\textit{irreducible})} + \underbrace{\mathbb{E}\big[(\mathbb{E}[m \mid z^\star]-g(z^\star))^2\big]}_{B_{\text{closure}}\ (\textit{decoder bias})}, \tag{2}$$

the cross term vanishing identically; (b) $B_{\text{closure}} \geq (\mathbb{E}[m]-\mathbb{E}[g])^2$*;* (c) $B_{\text{insuff}} \leq \mathbb{E}[\mathrm{Var}(m \mid \mathrm{bin}(g(z^\star)))]$ *for any binning of $g(z^\star)$; and* (d) B_{insuff} *is a functional of the law of $(m, z^\star)$ alone and does not involve g, so it is identical across closure conventions, while the bound in (c) is not.*

The proof is in §S2. Two of its parts are conditions of use rather than steps, and both are stated here. **The identity holds because $g(z^\star)$ is $\sigma(z^\star)$-measurable, and it fails if any argument of g is left out of the conditioning variable**: with $g = g(z^\star, T)$ and T excluded, the cross term need not vanish and the binning stops being a coarsening of the conditioning σ-algebra, which takes the identity and the upper bound with it. Where g reads a temperature (the broad IDAC set of §3.5.1, dim $=$ 103) we accordingly take $z^\star = (\text{profile pair}, T)$. On the VT-2005-matched set every row sits at 298 K and the two readings coincide. The Hammett closure of §3.6 reads a per-series constant pair as well as the substituent constant, so there $z^\star = (s, \sigma_{\mathrm{H}}^\star)$ and we condition on both.

B_{closure} is the systematic discrepancy of a fixed physical map in the sense of Kennedy–O'Hagan,[20,21] and its dominance over B_{insuff} is the empirical content of the claim that the ceiling is set by the closure rather than by the inputs. **What B_{closure} is the error of is a composite object.** The identity compares $g(z^\star)$ against $\mathbb{E}[m \mid z^\star]$, and $z^\star$ is produced by a procedure rather than measured: for σ-profiles, a quantum-chemical calculation on one conformer and one protonation state per molecule at one level of theory. B_{closure} is therefore a joint property of the map and of the procedure that produced the reference it reads, rather than of the kernel's functional form by itself. A displaced reference loads onto B_{closure} through the curvature of g even where g is exactly specified (§S8). The deployed pipeline consumes that same composite, so it is the object a practitioner faces, and every attribution below is to the pair rather than to the kernel. Where model-discrepancy calibration *fits* a correction from held-out observations, here g is fixed and $z^\star$ supplied from outside the fit, so we *attribute* the composite's error rather than correct it, and that is available only where an external latent oracle exists. **Exactness is contingent on g being fixed**: for a *fitted* decoder the cross term survives and a plug-in point split is invalid. Because B_{insuff} is bounded above and B_{closure} below, a positive *separation margin* $\mathrm{MSE} - 2B_{\text{insuff}}^{\text{up}}$ establishes that the closure's own error exceeds what input insufficiency can account for. A margin at or below zero is a failure to separate and licenses nothing: a reversal would need a *lower* bound on B_{insuff}, which lies beyond this instrument. **Its size is a lower bound and not an estimate.** Since $\mathrm{MSE} = B_{\text{insuff}} + B_{\text{closure}}$ exactly,

$B_{\text{closure}} - B_{\text{insuff}} = \text{MSE} - 2B_{\text{insuff}}$, and the reported margin under-states that difference by $2(B^{\text{up}}_{\text{insuff}} - B_{\text{insuff}}) \geq 0$, an amount beyond this design's reach. Every margin below therefore says the closure's error exceeds its inputs' by *at least* the number printed, and a bootstrap interval on a margin is an interval on that bound, its upper endpoint limiting nothing.

What population, and what sets the resolution. The population is the *measurement superpopulation*: a draw is a nominal compound pair *at a temperature*, together with the conformer, tautomer and protonation state actually populated there, and an independent determination of $\ln\gamma_2^\infty$ for it. The conditioning variable holds the reference profile and the temperature, leaving the conformational state and the replicate outside it, so $\text{Var}(m \mid z^\star)$ is the spread of $\ln\gamma_2^\infty$ across the conformational ensemble and across replicate determinations at one reference profile and temperature. That spread is strictly positive for physical reasons, whatever the sample size. Experimental label noise is a *component* of it rather than a term added beside it, and *observing* it is what the design forbids, each set holding at most one row at a given argument. The bounds are therefore conservative in one direction alone: were the superpopulation variance negligible, the finding would be the stronger $B_{\text{closure}} = \text{MSE}$. So over-estimating B_{insuff} can produce no artifact in what is reported, every entry is a bound rather than the share of the error the inputs contribute, and a low cluster-bootstrap frequency P is loss of resolution rather than evidence for the inputs. Resolution is governed by $\dim(z^\star)/n$.

2.1.2 The decomposition and the grounding gap

The decomposition is the measurement the conclusions rest on: assumption-light, and free of any need for a representable encoder or a lossless bottleneck. Alongside it we report a second, directly observable quantity, the *grounding gap* $\mathcal{G} := R_{\text{orc}} - R_{\text{e2e}}$, the oracle risk minus the best end-to-end risk. The paradox is $\mathcal{G} > 0$. Under a representable encoder (A1) and a lossless bottleneck (A2) it is sandwiched $0 \leq \mathcal{G} \leq B_{\text{closure}}$, so $\mathcal{G} > 0$ would certify $B_{\text{closure}} > 0$ (Theorem S1, §S2). A1 is unverified (the linear probe of §S6.1 does not bear on it, scoring an untrained encoder of the same architecture within its own spread), and A2 is unverifiable and physically dubious. When A2 fails, $\mathcal{G} > 0$ conflates closure misspecification with input insufficiency, a well-specified closure with a lossy latent giving $B_{\text{closure}} = 0$ yet $\mathcal{G} > 0$. We therefore read $\mathcal{G}$ as corroborating evidence and base the model-dominance conclusion on the decomposition.

The sign rule, and scope. Whether the *trained* physics model outperforms the *trained* black box at sample size n follows a sign rule.

Remark 1 (Sign of the physics penalty)**.** *Write* $\mathbb{E}[\hat{R}_{\text{phys}}(n)] \approx R_{\text{e2e}} + V_{\text{phys}}(n)$ *and* $\mathbb{E}[\hat{R}_{\text{direct}}(n)] \approx R_{\text{direct},\infty} + V_{\text{direct}}(n)$*, with* R_{e2e} *and* $R_{\text{direct},\infty}$ *the two asymptotic risks,* $V(n) \downarrow 0$ *the estimation variance of each estimator at budget* n*, so that the* physics penalty at budget n *is* $\mathcal{T}(n) \approx \Delta_\infty + [V_{\text{phys}}(n) - V_{\text{direct}}(n)]$ *with* $\Delta_\infty := R_{\text{e2e}} - R_{\text{direct},\infty}$ *its* asymptotic *part and* $D(n) := V_{\text{direct}}(n) - V_{\text{phys}}(n)$ *the variance a constrained prior saves. Then* (i) $\mathcal{T}(n) < 0$ *(physics helps) if and only if* $D(n) > \Delta_\infty$*: the variance a constrained prior saves must exceed its asymptotic bias.* (ii) *If* $\Delta_\infty > 0$ *and the variance gap vanishes (*$D(n) \to 0$*), there is a critical budget* $n^\star$ *beyond which* $\mathcal{T}(n) > 0$ *for all* $n > n^\star$*.*

Both clauses are rearrangements of the definitions above, and we state this as a remark for that reason: given the two expansions, *(i)* is an identity and *(ii)* its limit. What it supplies is vocabulary (a name for the asymptotic part, a name for what a constrained prior saves, and the budget at which they change places) rather than a prediction any measurement below is tested against. The limiting refinement in which $\Delta_\infty = B_{\text{closure}} - \mathcal{G}$ when the control is asymptotically Bayes goes beyond a rearrangement and is in §S2, with the working of both clauses. The consequence used below is that a prior is predicted to hurt precisely in the mature-data, low-noise, well-covered regime. Each physics-versus-control difference reported here is a penalty at the budget actually trained rather than an asymptotic one. We prove no "optimal grounding" theorem, and the decomposition leaves out-of-distribution accuracy where it was. Its variance term $D(n)$ is what carries the semiparametric phenomenon in which an estimated propensity outperforms the known-true one.[22,23] A fitted nuisance can be the more *efficient* estimator even when its bias is as large, so a true-valued input can miss the lower total risk at finite n.

2.2 Data curation: sources, splits, and labels

Solubility measurements come from BigSolDB 2.0:[24] 108,287 rows over 1525 solutes and 212 solvents. Melting points merged in as single-component crystal labels bring the modelled corpus to 127,930 rows over 20,766 solutes and 228 solvents between 243 and 438 K. We partition by Bemis–Murcko scaffold (a molecule's ring systems together with the linkers joining them, side chains removed) into 111,724 training, 8,103 validation and 8,103 test rows, so that every test solute is unseen at training. A melting-point-independent miscibility/structure filter sets row membership, and we merge on crystal T_m, ΔH_{fus}, Hansen parameters and infinite-dilution activity coefficients $\ln \gamma_2^{\infty}$ as per-solute labels where available. The rows of a split carry different labels: of the 8,103 test rows, 5608 carry a solubility measurement while the other 2495 carry a melting point alone, so every accuracy number below is on 5608 rows and says so (§2.4). 796 of those 5608 rows, 14.2% of them, carried by 26 of the 147 test solutes, have a solute of more than one component (salts, two ionic liquids and four hydrates). For those, the solid in equilibrium with the saturated solution differs from the one-component crystal Eq. 1's ideal term is written for, and all 26 of those solutes lack a measured melting point. They stay in: dropping the stratum moves the control's mean and the grounded arm's the same way and widens the physics penalty this paper reports, so removing it at this stage would flatter the comparison (§S3.2). Solid form is uncontrolled on both label sources, the solubility rows aggregating literature measurements on whatever form each laboratory had, and the $T_m/\Delta H_{\text{fus}}$ labels being merged from a separate compilation free to refer to another form. A polymorph difference therefore enters the ideal term as a systematic and the inter-laboratory disagreement below as part of its spread, and both documents leave it unseparated. Crystal supervision is scarce: the test and validation splits contain essentially no measured ΔH_{fus}, the quantitative basis of the identifiability argument (§2.1). The external single-component data we use for grounding are an open crystal-property pool (~15,000 melting points), the VT-2005 σ-profile database, and ThermoML infinite-dilution activity coefficients.[25] Two hazards the pipeline handles explicitly (melting-point unit detection, and the instability of the seeded scaffold split across pipeline versions) are recorded in §S8.

The label-noise floor, and the two functionals it is read with. Independent determinations of the same system disagree, and that disagreement floors what any predictor can achieve. On the raw BigSolDB rows, we group by (solute, solvent, temperature to 1 K) and keep the groups reporting at least two distinct sources. Within a group, the disagreement statistic is the mean absolute deviation of $\ln x_2$ from the group mean. Aggregating that per-group statistic over groups requires a choice of functional, and the two standard ones differ by a factor of five, so we report both. The mean over groups is 0.15 $\ln x_2$ units for the 3948 groups backed by two sources and 0.31 for the 349 backed by three. The median over the same groups is 0.03 and 0.09. Collapsing exact-duplicate values first (the same measurement re-tabulated across compilations, which would otherwise deflate the floor) leaves the mean at 0.16 over 4261 groups. The scaffold-split MAE reported here (1.70 to 1.85) is above the larger functional by a factor of five, so the accuracy ceiling stands clear of label noise under either reading. This floor is a mean absolute deviation in $\ln x_2$. The aleatoric estimate published

Table 1: Terminology and notation. Each term is defined in the text where it is first used; this table collects those definitions in one place.

Term	Meaning in this manuscript
closure g	the fixed, non-fitted physical map from the intermediate to the observable: here COSMO-SAC, and a Hammett relation in the second chemical domain
kernel	the segment-interaction energy inside COSMO-SAC; a closure variant is named for its kernel's year (*2002*, *2010*)
arm	one trained configuration, or one closure variant evaluated; *DirectGNN* is the closure-free control arm and *TGNNSolv* the closure-carrying one, abbreviated *TGNN* in Table 5
$z^\star$, $\hat{z}$	the closure's *entire* argument at its reference (tabulated) value, and the network's own prediction of it; a σ*-oracle* feeds $z^\star$ where $\hat{z}$ would go
grounding	two opposite operations on one reference database: *supervising* $\hat{z}$ against it during training, and *substituting* $z^\star$ for $\hat{z}$ at evaluation
B_{closure}, B_{insuff}	the error of the closure as deployed—the fixed map read at the tabulated intermediate—and what its inputs cannot resolve about the target: the two terms of Eq. (2)
margin	$\mathrm{MSE} - 2B^{\mathrm{up}}_{\mathrm{insuff}}$, with the MSE of Eq. (2)—the closure read at $z^\star$ against its own target m—so in squared m units, squared $\ln\gamma_2^\infty$ on the activity axis and not the solubility axis
binning bound	the upper bound $B^{\mathrm{up}}_{\mathrm{insuff}}$ of Lemma 1(c): the variance of m within bins of $g(z^\star)$, which fits nothing to the data
estimator cell	one setting of the binning bound's four analyst choices—bin count, bin edges, within-bin variance convention, combinatorial convention—together with the row or the pair as unit
$\mathcal{G}$	the grounding gap $R_{\mathrm{orc}} - R_{\mathrm{e2e}}$; distinct from Γ_S, the segment activity coefficient inside COSMO-SAC
stratum, map	a chemically defined row set of the broad IDAC set, and the 59 of them the estimator is run on
boundable	$n \geq 40$, so the fixed cell's eight bins hold five rows each
admissible	boundable *and* sign-stable under deletion of each contributing publication, in all four unit $\times$ convention cells
full, residual-only	which part of COSMO-SAC is evaluated, a model-form choice and not an analyst convention: *full* carries the Staverman–Guggenheim size-and-shape term, *residual-only* runs the segment-activity part alone and is what this pipeline deploys, its σ head predicting no cavity volume, so the term needs one from elsewhere
VT-2005, UD	the two σ-profile databases, never interchangeable: VT-2005 the Virginia Tech 2005 tabulation the σ head is supervised against and which is substituted for $\hat{z}$ at prediction time, UD the University of Delaware successor the activity-coefficient rows are matched to
labelled test rows	the 5608 of the 8103 scaffold-split test rows carrying a solubility measurement; the other 2495 carry a melting point only and are scorable in no arm
broad IDAC set	477 IDAC $\cap$ UD rows, 185 pairs, all temperatures, UD profiles
VT-2005-matched set	60 pairs at 298 K carrying a VT-2005 profile on both sides
σ	alone, the screening-charge-density profile; $\sigma(\cdot)$ with an argument, the generated σ-algebra; σ_{H}, the Hammett substituent constant, $\sigma^\star_{\mathrm{H}}$ its tabulated value and $\hat{\sigma}_{\mathrm{H}}$ the network's; σ_η, experimental label noise

for this task, an inter-laboratory RMSE of 0.75 in $\log_{10} S$ on 34 duplicate solutions,[1] is a different functional on a different base, and we keep the two apart below, pooling and converting neither into the other.

Disclosure: the scaffold guarantee is not certified for these runs. The released stream builder excludes any pool molecule whose scaffold appears in validation or test. One σ-stream build on disk was nevertheless generated against the wrong split files, and per-run stream provenance went unlogged, so which build fed which arm lies beyond the artifacts' reach and the guarantee is a property of the builder rather than of these runs. Of the 147 solutes spanned by the 5608 labelled test rows, 7 carry their own reference profile in that build, and 8, those 7 among them, carry a scaffold relative. The second count is therefore the union, and deleting it deletes both. The ungrounded arm carries no σ stream either, so the supervision contrast is the one a leak can reach and the 0.20 gain of §3.1 awaits leak-free certification from the re-run below. The file evidence, the contrast each fact bounds, and the one molecule the leak touches in the n=44 analysis of §3.3 are in §S8.

The leak-free re-run. We fixed its protocol before its numbers exist (§S8): the ungrounded and the grounded arm at five seeds (42 through 46), the published COSMO-SAC configuration unchanged, on a σ stream rebuilt against *this* split's files and certified inside each training run, read on the same 5608 labelled test rows and reported seed by seed. We fixed the decision it settles in the same pass, stating it in three branches rather than in the favourable one alone (gains of one sign, gains straddling zero, gains of the opposite sign), together with the list of displays the five-seed values replace. The supervision contrast is what that decision hangs on. All three branches leave the substitution result standing: it is one checkpoint evaluated two ways, beyond the reach of any σ-stream build.

2.3 Model, closure, and training

Our *shared* message-passing encoder embeds solute and solvent, a bipartite cross-attention block lets them interact, and the resulting pair representation feeds every head. We implemented two further backbones, and both stay outside the numbers reported here. The physics-channelled one was run against this encoder and trailed it, an observation we recorded with no set, seed count or interval attached (§S1), so we claim no accuracy comparison between backbones. A fusion head predicts T_m and ΔH_{fus}, giving the ideal term $\Phi(T)$ of Eq. (1). The activity term $\ln \gamma_2$ comes from one of three differentiable closures. Two are controls: the fitted NRTL (non-random two-liquid) excess-Gibbs-energy model, and its symmetric one-parameter collapse. The third, parameter-free, is the closure under study: a σ-profile head emitting a 51-bin screening-charge-density profile per molecule, and a COSMO-SAC layer mapping the profile pair to $\ln \gamma_2$. **That layer runs COSMO-SAC's residual, segment-activity part alone.** The Staverman–Guggenheim combinatorial term (the contribution molecular size and shape make to the entropy of mixing, and the second factor of $\gamma = \gamma^{\text{res}}\gamma^{\text{SG}}$ in Lin–Sandler) is off in every arm the substitution result is reported on. The σ head emits a profile shape and a cavity area while leaving the cavity volume unpredicted, so its own output falls short of evaluating the term. That is a model-form choice rather than an analyst convention: the kernel constants the layer uses are the 2002 values, regressed with the combinatorial term present, so the residual-only setting runs a fitted kernel outside its parameterisation. Its size on a published evaluation is a mean shift of -0.74 in $\ln \gamma_2^{\infty}$ over 7889 records, from $+0.03$ to -0.71 in bias. The two settings are the *full* versus *residual-only* pair on which §2.5 and §3.5 turn. One arm of Fig. 2 restores the term by taking a molar volume from a separate auxiliary head and passing it detached, so the combinatorial term is active while standing beyond the reach of the solubility loss. Inside the layer, the segment activities are themselves a fixed point, truncated at a count that is a free hyperparameter and *differs* between training and scoring: the COSMO-SAC arms here train at 16 segment iterations and are scored at 30. We chose sixteen against a convergence regression test on hand-built Gaussian σ-profiles, where the 16-against-30 gap is 6×10^{-5}. Hand-built profiles are the whole of that test's input, so agreement between the two counts on these arms stands unshown, and a checkpoint fitted at a different count is scored through an operator it was never fitted against. The exposure that carries to the substitution contrast is unmeasured. A second, independent truncation sits outside it: the SLE fixed point in x_2, run to a different count in training than in scoring. That second one is the one a deposit can re-solve, and the NRTL arm moves by $+0.0025$ in seed-mean MAE at its training count and a further $+0.017$ to convergence on the outer loop. That figure prices the outer loop rather than the segment one. Both truncations stay beyond re-solution on the COSMO-SAC arms, their kernel being a σ-profile pair that the deposit and the retained checkpoints alike omit (§S1). The saturated mole fraction solves $\ln x_2 = -\Phi(T) - \ln \gamma_2(x_2, T)$ as a damped fixed point in x_2, and a bounded, gated adaptive correction perturbs the physical parameters rather than the output, so the physical decomposition survives it.

External single-component data enter through *grounding streams* (the training-time sense of grounding). Each stream is a sidecar of *self-solvent* rows (no solubility label, only the target factor's mask active) interleaved into training under the guard above. It grounds the crystal factor from the melting-point pool and the activity factor from the σ-profile database, and so identifies from external data a factor that solubility alone leaves non-identifiable and that a black-box predictor is unable to use. Training runs a three-stage curriculum under a weighted loss, and the stages are named by number wherever one of them is referred to below: *P1*, a warm-up on the auxiliary streams alone; *P2*, joint training against solubility; *P3*, a low-rate stabilisation stage. The schedule pinned for the arms of Fig. 2 freezes the σ head at the end of P1, so the σ stream enters the warm-up only.

The in-house closures, and where their validation stops. The dispersion-off closures in the fidelity measurement (§3.5.3) are our own differentiable implementations, so those arms share one code path, and each consumes its own model-prescribed σ-averaging (Mullins for 2002, Hsieh for 2010). Our 2002 layer (the *2002, all channels* arm wherever the kernels are compared, so named because it sums the segment contributions of all three profile channels rather than the nonpolar one alone) reproduces the NIST reference COSMO-SAC-2002[26] to RMSE 0.0035 $\ln \gamma$ on identical VT-2005 profiles. Our COSMO-SAC-2010/dsp layer,[27,28] built on the *typed* latent of three donor/acceptor-resolved σ-profiles per molecule in place of one, reproduces the NIST 2010 reference with dispersion *off* to RMSE 1.6×10^{-4} $\ln \gamma$ over 400 pairs, the largest single departure being 1.7×10^{-3} (§S1). The dispersion term is *not* validated to that standard, so we compute every dispersion-off contrast on our layers and quote the *2010, dispersion on* runs from `cCOSMO`, that reference implementation itself.

The same closure against a published evaluation. Reproducing a reference implementation fixes the code, and it leaves open whether the error this paper decomposes is the error a published COSMO-SAC has or something peculiar to this data assembly. Table 2 answers that. De Souza et al.[29] score the original Lin–Sandler COSMO-SAC on the infinite-dilution activity-coefficient database distributed with *The Properties of Gases and Liquids*, at AAD $= \mathbb{E}|\ln \gamma^{\infty}_{\mathrm{calc}} - \ln \gamma^{\infty}_{\mathrm{exp}}|$ of 1.75 with HF-TZVP profiles and 0.68 with BP-TZVPD-FINE: one closure, two profile databases, a factor of 2.6 between them. An evaluation of COSMO-SAC is in that sense an evaluation of the profiles it was handed, which is why B_{closure} is defined above on the composite of map and profile calculation and not on the kernel. We ran our own differentiable layer over the same records (§S8), and on AAD it lands inside the bracket the published closure spans across profile databases, under both conventions. On R^2 it is inside in the residual-only convention and level with the bracket's lower end in the full one, the two printing alike at 0.88. Our own two sets are inside on AAD alone, and in the residual-only convention only: their R^2, 0.61 and 0.40, is below the published range. Two limits stay open. The profile database is the one axis we cannot match, VT-2005 being what this paper deploys and the axis that publication shows dominates. And whether the publication's 6977 points are the same rows as the 7889 scored here is unproven, the record selection not being stated file by file. Subject to those, the closure error decomposed in §3.5 is, in AAD, the error a published COSMO-SAC has. Which half of this paper's chemical localisation carries across to that set, on 130 times the data and someone else's compilation, and which half does not, is in §3.5.3. The same publication scores a later kernel on the same records, and its rows are in the table because they put the deployed one in its place: COSMO-SAC-HB2, which types segments as donors or acceptors, reaches AAD 0.46 to 0.48 at R^2 0.96 to 0.97, about half the deployed closure's 0.77 and a third of the 1.75 that anchors the profile-database bracket. We run the 2002 kernel because it is the one whose differentiable re-implementation can be checked against a public reference here, and the results below are substitutions inside a fixed map rather than claims that the map is the best available. §3.5.3 runs the 2002 → 2010 step on our own sets and finds the improvement confined to the smaller set's chemistry. Every ceiling this paper reports is accordingly this kernel's ceiling and not COSMO-SAC's.

2.4 Evaluation protocol, and the DirectGNN control

In the *oracle substitution* we replace the predicted σ-profile $\hat{z}$ with the reference VT-2005 profile $z^\star$ before the closure, matching by canonical SMILES. One object is replaced, not two: the σ-head emits a normalised 51-bin shape and a cavity area whose product is the area-weighted profile the closure reads, and the tabulated pair stands in the same relation, so the reference area reaches the residual term's A_2/a_{eff} prefactor as part of that single replacement (§S1).

A property of our head bears on what that substitution does, and it is a defect rather than a choice. The encoder carries a separate learned adapter for the solute and the solvent role, and the σ-head reads whichever the molecule occupies, so one molecule has *two* predicted profiles. A σ-profile is a property of a molecule alone, and the closure reads the pair, so the model can and does violate an identity no activity model may violate. On a self-pair, $\ln\gamma_2$ must be exactly zero. We fed every one of the 2634 distinct test solutes to both roles at its own temperature. Of those, 102 are untestable, the solver sitting at $x_2 = 1$ there, where the residual term vanishes identically, so $\ln\gamma_2$ is zero by composition rather than by profile agreement. On the 2532 that are testable, the median $|\ln\gamma_2|$ is 0.154, the mean 0.286 and the largest 5.78, with 95% above 0.01. With one profile on both sides, the same computation returns zero to machine precision, so the role gap is the whole of the violation. Which of the readout's three numbers carries that gap decides how large the repair is, and on these arms it is the profile: with the two roles agreeing on the shape the residual bracket is identically zero whatever the areas, so on a self-pair the profile disagreement is the whole of the violation (§S1, which also prices a mismatch on the area and on the volume, and gives how far the deployed arms' areas sit from the tabulated ones). The substitution overwrites both roles at once and therefore closes that gap as well. That is a second thing it changes besides the profile values, and we leave the two unseparated; tying the roles is the repair, and it is a retraining. We match solute and solvent independently. The substitution replaces those two profiles and stops there, the crystal head reading a solute-only embedding it leaves untouched. The bounded correction does read the closure's own parameters, so the crystal values the solver finally receives shift by a little: pooled over the five seeds, at most 3.2×10^{-5} in Φ against a median $|\Phi|$ of 4.5, on the 8066 of the 8103 test rows where the substitution applied and on the 5608 labelled rows alike (§S1). The reference table covers 99.3% of the labelled test rows on the solvent side and 5.3% on the solute side, so the solvent profile carries the substitution contrast below.

We apply it three ways to rule out implementation artifacts, differing in *when* the replacement is made: at evaluation alone; injected at training so the model co-adapts; and a channel swap, injected at training under a coordinate-descent schedule that freezes the crystal branch during the SLE phase so the activity branch alone refits against it. The last two are at seed 42. An analogous override substitutes reference crystal targets. The activity target for the decomposition is the infinite-dilution $\ln\gamma_2^\infty = \lim_{x_2\to 0}\ln\gamma_2$, on two sets that differ in their reference database and their regime (§3.5.1–§3.5.2).

DirectGNN shares the encoder, interaction, readout and pair representation, adds a thermometer (cumulative-threshold) temperature encoding, and maps directly to $\ln x_2$ with no

Table 2: The deployed closure against a published evaluation of the same closure, and this paper's own two sets through it. AAD is the mean absolute error in $\ln \gamma^\infty$, n the number of scored records.

COSMO-SAC-2002 on	profiles	n	AAD	R^2
As published[29]				
PGL 6th ed. IDAC set	HF-TZVP	6977	1.75	0.88
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.68	0.96
A later kernel on the same evaluation[29]: COSMO-SAC-HB2[b]				
PGL 6th ed. IDAC set	HF-TZVP	6977	0.48	0.96
PGL 6th ed. IDAC set	BP-TZVP	6977	0.48	0.97
PGL 6th ed. IDAC set	BP-TZVPD-FINE	6977	0.46	0.97
The same set, through this paper's closure				
full convention[a]	VT-2005	7889	0.98	0.88
residual-only	VT-2005	7889	0.77	0.92
non-aqueous rows	VT-2005	4347	0.47	0.68
aqueous rows	VT-2005	3542	1.13	0.83
This paper's own sets, residual-only convention				
broad IDAC set	UD	477	1.12	0.61
VT-2005-matched set	VT-2005	60	0.81	0.40

[a] The like-for-like row, a published COSMO-SAC carrying its Staverman–Guggenheim term. The residual row is the setting this paper's B_{closure} is measured in.
[b] A donor/acceptor-typed hydrogen-bonding kernel, rather than the kernel this paper deploys. Printed because it is on the same 6977 records and beats the deployed closure by a factor of about two.

heads, closure or solver, so it sees T explicitly where the physics arm sees it through $\Phi(T)$ and the closure. We tuned the optimisation settings per arm, and they differ by up to an order of magnitude. The TGNNSolv−DirectGNN gap is therefore a comparison of two separately tuned pipelines that differ in the thermodynamic bottleneck, and not an isolation of it; every physics-penalty comparison below carries that caveat (§3.7).

We report the controlled comparisons of §3.1 as mean±sd on a fixed row set, at five seeds for the arms the leak-free re-run trains and three for the rest. That lock is the 5608 test rows carrying a solubility measurement, 5608 of the 8103 (§2.2), so all arms score identical rows. The 2495 rows outside it carry a melting point alone and are unscorable in every arm alike, the control included, each dropped for its label and not for anything an arm predicted. Which rows carry a label fixes the set: the lock's finite-prediction clause is inert, removing no row from any arm at any seed, and §S3.2 gives the decomposition. The runs behind every number reported here fall into twelve families, a family being a set of runs sharing a script, a corpus and a seed policy, and Table S25 gives each number its family, its seed count and the artifact it regenerates from.

2.5 Estimating the split: one cell, one convention, one rule

The instrument we fix in this section is a procedure, and it takes three inputs and nothing else: a fixed closure g, a table of reference values $z^\star$ its argument can be read at, and measurements m of its target on rows where both exist. It returns, per chemically defined row set, the closure's own error MSE, an upper bound on what the inputs leave unresolved, the margin between them, a cluster interval on that margin, and a verdict of admissible or otherwise with the test that decided it. Four analyst choices set the binning bound of Lemma 1(c) (the bin count, where the equal-count bin edges fall, the within-bin variance convention, and the combinatorial convention), and we fix all four here, before any margin is reported. One setting of them, together with the choice of the row or the molecule pair as the unit, is an *estimator cell*. The four rules below are the procedure in full. The map of §3.5.1 is what running it on this closure returns.

The estimator cell. Eight equal-count bins of $g(z^\star)$, the *unbiased* (Bessel-corrected) within-bin variance, and the row unit, applied identically to every set and every stratum whatever its size. The bin count is the choice with the most leverage (it takes the broad set's aggregate margin from +0.01 to +0.96), so the cell never stands alone anywhere a finding rests on it: we run the same sweep for the strata the admissibility rule passes, and report it with them. Enumerating every equal-count contiguous partition, the bin-edge choice is immaterial on the broad IDAC set and material on the VT-2005-matched set, whose margin changes sign inside the envelope. We accordingly read that set's margin as consistent with zero and carry the claim on the broad set in stratified form.

The convention rule. The margin depends on the combinatorial convention, so one rule governs both sets: every margin is read in the *residual-only* convention, pairing MSE and the bound computed under that same convention. Taking the residual-only convention on the smaller set and the full convention on the broad one would be the favourable choice on each set taken separately. One rule applied to both costs the broad set's margin about half its value. The Jensen constant-offset bound is convention-specific (it separates under the full convention and inverts under the residual-only default), so what we report rests on the binning bound, which fits nothing and therefore leaves every held-out row to stand on its own rather than borrowing strength from a near-twin. Both bounds are valid on the one population quantity, by Lemma 1(d), and so is the smaller.

The stratification rule. The broad set is stratifiable where the VT-2005-matched set's 60 pairs resist stratification, on two axes that are pure functions of molecular structure: an ordered nine-class cascade of substructure tests over the solvent, hydrogen-bonding role first and dominant functional group second, and the solute's role as water or as an organic. Every class name is a substructure test rather than a chemical judgement: a *glycol ether* here is a solvent carrying a hydroxyl group and two or more oxygens, and an *aprotic* solvent one lacking an O–H or N–H hydrogen to donate. The classifier reads a SMILES string and stops there, leaving m, g, the squared error, the source and the bootstrap unread, so every cell was fixed before its outcome, which is a property of the code rather than an assurance. A stratum is *boundable* at $n \geq 40$, where eight bins hold five rows. We report the smaller ones at that same cell, with the adaptive bin count $\max(3, \lfloor n/10 \rfloor)$ in the rows n it is applied to carried beside them in the deposited table, nothing being dropped for being small.

The admissibility rule. A stratum is *admissible* only if it is *(a)* boundable at that fixed cell and *(b)* robust to leave-one-source-out (its margin keeps its sign when each source publication contributing to it is deleted in turn) in all four unit $\times$ convention cells, since the map reports all four. That last clause is load-bearing and we keep it: drop it and the set taken whole passes too. A stratum drawing on one publication fails (b) by non-testability, there being no deletion to perform. A deletion leaving fewer than 16 rows leaves the fixed cell undefined, and fails (b) as well. A third clause reduces the admissible *cells* to row sets, and it is stated here with the same precision rather than left to prose, because the one claim this paper carries from the map and the multiplicity null that tests that claim both turn on it. *(c)* Admissible cells holding identical rows count once. An admissible positive row set X is *demoted* to another, Y, when Y is admissible and positive, Y holds more rows than X, more than half of X's rows are Y's, and the residual $X \setminus Y$ falls short of admissible, so that X's margin stays inside Y. That residual decides it rather than the overlap: a row set most of whose rows lie inside a larger one survives (c) whenever its own margin stands outside that larger one, so *(c)* tests margins rather than containment, and overlap by itself demotes no cell (§S3.8). Clause (c) takes the six admissible cells of §3.5.1 to three distinct row sets. Positivity then removes one of the three and (c)'s residual test the second, leaving one. It is rebuilt on every draw of the multiplicity null, on the permuted labels, like the rest of the rule (§S3.6). Everything else is *reported in the map and not stated*, with its n, its sources, its margin, its interval and the reason. The rule governs the cells that favour us and those that count against us alike, and it tests whether a number is stable rather than what it means: an admissible stratum whose margin sits at or below zero, and one that (c) demotes, both establish nothing.

3 Results and discussion

A cluster-bootstrap frequency P is the frequency, over two-way (solute $\times$ solvent) cluster resamples, with which the margin it is attached to stays positive. It is a resampling frequency rather than a p-value against any null. We quote such frequencies as bands (rounded to one decimal, with > 0.95 and < 0.2 closing the ends), because with 17 solvent and 41 solute clusters on the VT-2005-matched set they carry at best one decimal of resolution. The deposited artifacts hold the two-decimal values. Our resampling intervals (the ranking and recalibration intervals of §3.2, the margins of §3.5 and the pKa intervals of §3.6) are *percentile* intervals: their endpoints are order statistics of a resampling distribution, uncorrected for bias and unstudentised, over the unit named beside each. §3.2 resamples the 107 solute clusters, 4000 draws, at the 95% level. §3.5 resamples solute and solvent clusters two-way, 3000 draws. In §3.6 the decomposition bounds resample molecules within each Hammett reaction series and the flip gap resamples the four series themselves, 4000 draws. Some $\pm$ and bracketed spans below are of a different kind, two in particular: the kernel-transfer figures of §3.5.3 are a standard *error* over five fit seeds, and the leave-one-publication-out span of §3.5.1 is the range of that curve. Every claim below carries its set, its size and its scope in the sentence that makes it, and §S3.1 collects the same nineteen side by side as one table (Table S2).

3.1 Supervising the intermediate toward the reference, and substituting it at prediction time

The word "grounding" names two opposite operations on the same VT-2005 database. Every solubility figure in this subsection and the next is an $\ln x_2$ MAE over one row set, the 5608 of the 8103 scaffold-split test rows that carry a solubility measurement (Table 1). The single-seed control arms below carry that identical set, so every comparison here stays inside one row set. Supervising the σ head against the database *during training* moves solubility MAE from 2.037±0.064 (ungrounded) to 1.927±0.094 (grounded) over five leak-free seeds, a mean of +0.11 that is not a sign: the per-seed gains are +0.149, +0.122, +0.175, −0.017 and +0.122, and the two arms' per-seed values interleave, the grounded arm the worse of the two at seed 45. The two arms differ in training budget: the grounded arm precedes the shared curriculum with a σ warm-up on the grounding pool of up to 40 epochs, early-stopped on a held-out tenth, while the ungrounded arm sets that budget to zero (§S3.2). The warm-up against permuted σ targets that would separate the two was left unrun, so what the gain prices is the intervention entire, extra pretraining and accurate targets together, rather than the accuracy of the reference values by itself. Those spreads are over initialisation and batch ordering, so they price the training run rather than the draw of test solutes. Pooling the five seeds and resampling the 147 test solute clusters, we put the gain at +0.110, 95% percentile interval $[+0.001, +0.222]$. The two prices hold different things fixed and they do not agree: over the draw of solutes the interval clears zero at its endpoint, and over training runs the gain has no sign at all. Since the ungrounded arm carries no σ stream, this is the one contrast in the paper a σ-stream leak could inflate, and it is what the re-run was pre-committed to settle. Its stream is certified inside each training run and reaches no test molecule: 0 by canonical SMILES against 5101 held-out SMILES, and 0 by Bemis–Murcko scaffold against 2702 held-out scaffolds (§S8), so on these arms there is nothing to delete. The published three-seed family, whose stream build is uncertified, put the same gain at +0.198 $[+0.06, +0.33]$, and at +0.172 $[+0.03, +0.31]$ with the eight leak-reachable solutes deleted, on 351 of the 5608 rows. That pair bounds where a leak can act in *that* family and is what limitation (xii) refers to; it is superseded here by a build in which a leak cannot act at all. Substituting the reference profile for the learned one *at inference* degrades it, and that substitution alone is the effect this paper is named for. It erases any positive R^2 (Fig. 2): the model predicts solubility better *from its own learned σ-profile* than when *given the external reference one*, at MAE 1.93 against 2.34 in $\ln x_2$ units, over the same five seeds and again with the two arms' per-seed values disjoint. Resampling the same 147 solute clusters over the pooled five seeds, we put that penalty at +0.408, 95% percentile interval $[+0.29, +0.54]$. The median per-row gap is +0.30 against the mean's +0.41, so the effect sits at the middle of the distribution and grows heavier above it. Of the figure's eight arms, two are non-COSMO controls, DirectGNN and a fitted NRTL closure. Five COSMO-SAC arms differ in how the σ head is trained (the warm-up asymmetry above included) and in what it is fed, and the sixth (*grounded+comb.*) differs instead in the closure, restoring the Staverman–Guggenheim combinatorial term. *Ref. σ (eval)* is the grounded checkpoint re-evaluated with the reference profile substituted at test time, and *ref. σ (train)* the co-adaptation control injecting it during training instead.

That +0.41 is the evaluation-only substitution, and part of it may be unrelated to the closure: swapping an input distribution at test time degrades a pipeline on its own. The control that removes that component injects the reference profile *at training*, so the model co-adapts. It was run once, at seed 42, in the published family and not in the re-run, so it admits two readings and neither is a matched five-seed contrast. Against its own family's seed-42 learned arm (1.803) it costs +0.18, to 1.981, where the evaluation-only substitution costs +0.43 at that seed (2.232); against the re-run's seed-42 learned arm (1.887) the same injected value costs +0.09 and the evaluation-only substitution +0.26 (2.145). A channel-swap control, seed 42 and of the published family as well, reaches 2.08: +0.27 under the first reading and +0.19 under the second. What co-adaptation carries off is a share of the substitution gap, and we withdraw that share as a magnitude at any seed. The learned arm's between-seed range is 0.238 at five seeds, the injected arm was measured once, and two of the four recomputations that hold it at its measured value while the other two arms take another seed's exceed one. A share above unity is a construction that has stopped meaning what it is named rather than a wide interval (§S3). Both controls sit above the learned arm under both readings, which rules out an implementation artifact and is the whole of what they establish. It carries a second confound: whether either trained on the same σ-stream build as the comparator it is read against is uncertified, and the file times leave it open either way (§S8). The five-seed arms are the ones that can be set against one another, and there the evaluation-only substitution costs several times what supervision gains (+0.41 against −0.11). That ratio is between arms that carry the co-adaptation confound, so it measures something other than the paradox's co-adaptation-free magnitude. The arm that supplies that magnitude, at +0.09, was run at one seed and stands outside any ordering against the −0.11, whose own per-seed values (+0.149, +0.122, +0.175, −0.017, +0.122) straddle zero rather than straddling it. Which of the two operations is larger is therefore unresolved, and only their signs are carried forward.

3.2 What the substitution costs: level, spread, and the choice of solvent

Figure 3 sets the same rows out predicted against measured, for the control and three of the closure arms. Two things are visible there that the arm ordering hides. First, every arm is severely shrunk toward the corpus

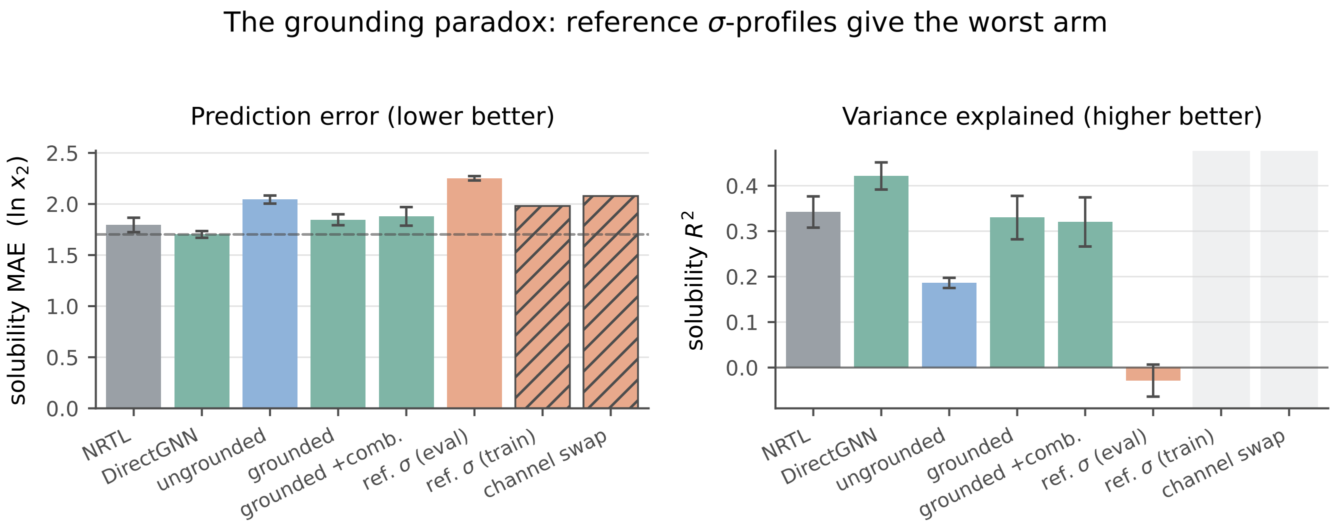

Figure 2: Controlled comparison on the 5608 labelled test rows of the scaffold split. Solid bars: five seeds, mean ± a population standard deviation over them. The ungrounded, grounded and σ-oracle arms come from the five-seed leak-free re-run. The two hatched bars are the co-adaptation controls, seed 42 only, so they are drawn without an error bar or an R^2, keeping an empty slot in the right panel. Dashed in the left panel alone, the DirectGNN reference.

mean. The ordinary least-squares slope of predicted on measured is 0.515 for DirectGNN, 0.399 ungrounded, 0.511 grounded and 0.658 with the reference profile, all of them well short of 1. In every panel the median trace crosses the identity: sparingly soluble compounds are called more soluble than they are and highly soluble ones less. All four slopes are seed 42's, the figure's seed, with one run behind each. That shrinkage originates outside the closure: at that seed the closure-free control's slope differs from the grounded closure's by 0.004, so whatever produces it lives in the encoder and the data. Second, the reference-profile arm is the *least* shrunk and the worst. Its slope is the closest of the four to 1 and its prediction spread exceeds the measurement's, yet its MAE is the highest and its R^2 is indistinguishable from zero. Substituting the reference restores spread that is misaligned with the truth, and its worst region is the sparingly-soluble tail it appears to reach. That tail is where the closure is driven hardest. At seed 42 the substituted arm returns 119 rows with $\ln \gamma_2 > 10$ where the learned arm returns none, the learned arm's largest being 7.77 against the substituted arm's 14.6. And 141 of its $\ln x_2$ predictions fall below -15, to a minimum of -22.3 where the corpus itself stops at -17.7. The worst 500 of the 5608 rows carry 0.25 of that seed's +0.26 mean gap. We score both arms at the 30 segment iterations of §2.3, with the fixed point left short of convergence there, so those rows are the ones on which the truncation is least certified as well as the ones on which the two profiles differ most. What the count is worth on these arms lies beyond the retained artifacts, which keep none of their weights. The tree does keep one trained COSMO-SAC head (a retired run whose weights were fit at 8 iterations, and which scores 2.61 at 30 against 1.92 at 8 on these same labelled rows). On it the substitution penalty moves with the count as well: substituting the reference profile costs $+4.32$ $[+3.61, +5.25]$ at 8 against $+6.43$ $[+5.39, +7.68]$ at 30 over 5571 rows in 64 solvent clusters whose solvent carries one. That head departs from the penalty reported here in both size and direction, its own being fifteen times larger at the count the +0.41 is measured at and moving its predictions the other way, so those two figures price the lever rather than this paper's +0.41. That is the misspecified map made visible on the solubility axis.

The substitution also costs the decision the model is used for. The separate-tuning confound of §3.7 leaves this comparison clear: the reference-profile arm re-runs the grounded checkpoint's own weights over the same rows and changes exactly one thing, the profile fed to the closure, so one tuning serves both sides. They differ in Φ by at most 3.2×10^{-5} over these 5608 rows at the worst of the five seeds. Co-adaptation is the confound that survives. Like the MAE gap above, this is an input swapped at evaluation; we ranked no arm that had trained on the reference profile, and what co-adaptation would do to an ordering stays unmeasured here. We group the labelled test rows by solute and temperature and rank the solvents inside each group (823 groups over 107 solutes; Table S19). The mean per-solute Spearman correlation then falls from 0.474/0.612/0.596/0.515/0.556 to 0.361/0.399/0.365/0.343/0.362 across the five seeds, pooled -0.185 $[-0.27, -0.11]$, every per-seed interval excluding zero. Four of the five falls exceed any difference between seed replicates of a single arm on this measure, the largest of which is 0.132; the fifth, at seed 42, is 0.114 and does not, so at that seed the fall is inside the spread a single arm shows between its own seeds. The deposit carries that benchmark for the rank correlation, so we read it against that measure and leave the two below unbenchmarked. Every interval here is a 95% percentile bootstrap over those 107 solute clusters, the effective sample rather than the 823 groups. Two decimals is the precision the draw count determines, coarser than the precision the deposit records: at the 4000 draws behind them a third decimal is a property of the bootstrap seed, so we apply to these the coarsening rule §S3.8 states for the map. Normalised discounted cumulative gain over the top three solvents (nDCG@3) falls by -0.071 $[-0.11, -0.03]$, and top-1 accuracy by -0.088 $[-0.17, -0.01]$. We quote that last figure pooled, and pooled only: three of its five per-seed intervals span zero, so of the three ranking metrics it is the one whose sign the per-seed columns leave open. The solvent the model would select changes in 42 to 53% of groups, asymmetrically: the learned profile by itself picks the measured best solvent in 18.2/22.1/24.2/21.4/19.6% of groups, the reference profile by itself in 13.2/10.9/12.5/14.7/10.0%. All of that is an ordering rather than a level. Fitting a separate affine recalibration inside every group (a slope as well as an intercept, the largest concession a miscalibration reading can ask for) removes 91% of the MAE gap, $+0.426$ $[+0.30, +0.55]$ to $+0.040$ $[-0.00, +0.08]$, and what is left of it no longer stands clear of zero. That is the restored, misaligned spread of the paragraph above in another currency. That map is fitted on the rows it is then scored on, two free parameters against a median of five solvents, so it bounds what the concession can buy rather than measuring it. Refitted leaving out the row it scores, it returns $+0.32$ $[-0.61, +1.20]$ pooled, an interval wide enough to hold both the whole gap and none of it. On the 550 groups that hold at least five solvents, where a refit has something to work with, it gives +0.17/+0.14/+0.28/+0.23/+0.19 by seed, three of the five intervals spanning zero (§S6.1). The MAE penalty is level and scale in the sense that a per-group map fitted on the answer absorbs most of it. Out of sample the data support no statement in either direction. The ranking penalty survives both fits, by invariance rather than by any centring argument. That invariance is to an *increasing* affine map, and the fitted map decreases in 10 to 15% of learned-profile groups and 15 to 21% of reference-profile ones, where it reverses that group's order outright. Restricted to the 485 groups of 823 whose map increases in both arms at all five seeds, the ranking gap is ρ -0.107 $[-0.179, -0.031]$, with nDCG@3 and top-1 then spanning zero. Only the rank correlation survives that restriction, and the subset it survives on is 59% of the groups.

What stands in its place is indistinguishable here from a solute-blind ordering. Two references are blind to the solute: the same arm's own predictions for a randomly drawn other solute over the same solvents, and a ranking of those solvents by their leave-one-solute-out mean measured solubility. Set against both, the learned-profile arm is above them on Spearman, Kendall, top-1 and nDCG@3 at every seed: all forty of its margins are positive, and 27 of the forty intervals exclude zero. The thirteen that do not are concentrated rather than scattered, seven of them at seed 42 and four at seed 45, so the separation is a property of three of the five runs and not of every one. The reference-profile arm's own forty margins are smaller, 36 of them positive, and their intervals carry the reading: 36 of the forty include zero, and the four that do not are all one seed's and one reference's, the donor floor at seeds 43 and 44 (Table S19). That measures a failure to separate the reference-profile arm from solute-blind orderings on most of the design, and it stops short of demonstrating that the two are inseparable. Both arms are above chance, the within-group permutation floor being $\rho = 0.00 \pm 0.02$ for each. Supplying the true physical intermediate costs an ordering rather than a calibration: the arm's ordering of solutes becomes indistinguishable from one drawn blind to them.

3.3 What moved inside the model, and whether a downstream correction repairs it

A candidate *mechanism* for the low-fidelity, end-to-end regime runs as follows. A σ-profile head trained *end-to-end* through a fixed, misspecified closure (with no σ supervision in that phase) is under pressure to *cancel* the closure's error rather than to be physically faithful. The hypothesis has two parts, and this section establishes the first of them. Its antecedent, that the latent leaves the reference along a direction shared across molecules rather than as per-molecule noise, is measured. Its consequent, that the departure cancels part of B_{closure}, stays unmeasured, and the one direct check available argues against it. The external reference that fixes the ceiling makes the first part testable: we compare the predicted profile $\hat{\sigma}$ against the reference VT-2005 profile molecule by molecule (n=44). That comparison is the anatomy of the object the two operations of §3.1 act on, namely how far the learned intermediate has travelled from the reference that supervises it and then replaces it. We place it here, inside the phenomenon, for that reason rather than as evidence for why either operation has the sign it does.

What is measured, and on what. We make the comparison within a single grounded model, checkpointed before and after solubility fine-tuning, over the n=44 matched molecules, and repeat it at three seeds. Those three runs are a *separate* set from the arms that produce the paradox, trained with the σ head left unfrozen through fine-tuning, where the four COSMO-SAC arms of §3.1 hold the head from P2 onward. That freeze holds the head while leaving the encoder that feeds it free, so those arms' profiles stay free to move from their warm-up value too. How far *their* latent drifted is unmeasured and lies beyond recovery from the deposit. What follows is therefore measured on runs of its own and stays unestablished of the checkpoints whose paradox it is offered to explain. Under σ supervision the head reproduces the reference profile to within $36\pm1\%$, so the intermediate *can* carry the physical shape, though only as an in-sample fit, all 44 probe molecules lying inside the σ-grounding pool. Solubility fine-tuning then drives it a further $41 \pm 3\%$ away, for a total departure from the reference of $51 \pm 2\%$, a magnitude reported but unstable across analysis configurations, so what these runs fix is the sign and the shared direction below rather than the size. A relative deviation carries no scale of its own. The ladder that supplies one (§S5: the same 44 molecules and the same relative-L_2 metric, its rungs built from the reference profiles the grounded arm was supervised on) places the fine-tuned latent at 0.507 and the grounded base at 0.364. Two further rungs set the scale: 0.493 for one corpus-mean profile reused for every molecule, a rung carrying no molecular information at all, and 0.651–0.684 for each molecule given another molecule's reference profile. Read on that ladder, the drift consumes essentially all the molecular specificity grounding had provided, which is what makes the shrinkage reading below more than a formality.

What the departure is, and what it is not. We find the drift structured rather than

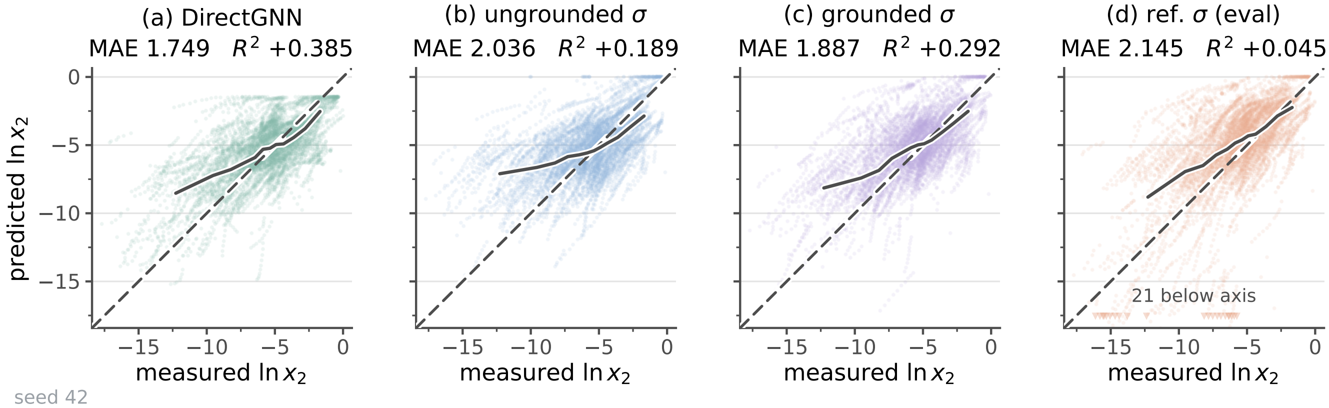

Figure 3: Solubility parity for four arms of the controlled comparison: predicted against measured $\ln x_2$ on the 5608 labelled test rows of the scaffold split, which are 763 solute–solvent pairs over 147 solutes and 70 solvents measured between 248 and 368 K. (a) DirectGNN, the closure-free control; (b) COSMO-SAC over an unsupervised σ-profile; (c) the same closure with σ-profile supervision during training; (d) the arm of (c) with the reference VT-2005 profile substituted at evaluation. Dashed, the identity. Solid, the median prediction in twelve equal-count bins of the measurement. Each panel header carries that panel's own MAE and R^2. Both axes span the measured range with a margin around it, and the markers along the foot of (d) are its 21 predictions that fall below the frame, each at its own measured value. Panels (b), (c) and (d) come from the five-seed leak-free re-run: the ungrounded, grounded and σ-oracle arms respectively.

per-molecule noise, and its structure goes beyond a constant offset. The first two principal components of the mean-centred deviation carry $72 \pm 1\%$ of the variance. The quantity we read against a null (the top-two share of the learned-versus-reference deviation, 0.399) clears a phase-randomised null (0.221, $p < 0.001$), on one grounded checkpoint rather than over the three seeds. Its 0.221 carries a spread of ± 0.01 that is the null's own over draws rather than a spread over seeds. Two nulls it fails to clear travel with that. The mean-centred share has no matched null of its own, and a wrong-molecule null built from real reference profiles does *not* clear (0.772, $p = 1.00$), so a reader who declines the case for preferring the phase-randomised null should read the spectrum as uninformative rather than as supporting. One distortion operator fitted on half the 44 molecules predicts the drift on the other half $4.1 \pm 0.5\times$ better than the zero-drift null, the $\pm$ spanning the three seeds at one partition of the 44 rather than the partitions. That null is a deliberately weak one and is uncontrolled on these three seeds: a molecule-independent constant offset already reaches $2.1\times$ on the one companion run where that control can be computed. So the ratio stands against the identity null alone, and structure *beyond* an offset rests on the mean-centred spectrum. The held-out half is also homologue-saturated (six n-alkanes, four halobenzenes, five pyridines and four butyl and hexyl amines among the 44, and the median held-out molecule sits 16% of its own profile norm from its nearest training-half neighbour), so "outside the fit" is largely within-cluster interpolation. That spectrum leaves open a shrinkage of every profile toward the corpus-mean shape, which is how the drift looks on the reference ladder of §S5. The dominant mode (Fig. S4) moves area out of the nonpolar bins and into the polar wings: fine-tuning makes the network represent molecules as more polar than its *own grounded profile* does. The plotted drift is taken against that profile rather than the reference, and the mode is defined up to sign, drawn in the mean drift's direction.

Whether the departure cancels closure error. It remains unshown, and the one direct check runs against it. Holding the cavity area at its reference value, the sole retained end-to-end SLE-trained COSMO-SAC checkpoint scores 18.75 against 1.47 on the n=60 VT-2005-matched pairs: on the activity axis its drifted profile is an order of magnitude *worse* through the same closure. Four limits sit on that number. The checkpoint is the uncontrolled one, whose 82% departure is half again the three-seed 51 ± 2%. It is a retired run, trained at 8 segment iterations and scored at 30, and its own scaffold-split accuracy is MAE 2.61 at $R^2 = -0.31$, worse than every arm of Fig. 2. The axis is wrong, the drift being produced at finite composition and scored at infinite dilution. And the check has almost no power in the direction that matters, so it separates "no cancellation" from "large cancellation" while leaving "some cancellation" open. The kernel sweep reported beside it falls short of a second check in the same direction, and we state it here for what it does bound. Sweeping the closure's hydrogen-bond coefficient at zero, one and two times its standard value over those same n=60 pairs gives mean squared errors 2.42/2.83/10.15 on the strong-donor pairs, 0.090/0.092/0.918 on the acceptor-only ones and 1.205/0.867/0.578 on the inert ones. The three classes separate by direction rather than by size: both hydrogen-bonding classes rise with the coefficient (the acceptor-only pairs by tenfold and the strong donors by fourfold) while the inert class falls monotonically by half. That is consistent with B_{closure} sitting in the hydrogen-bond kernel, and it stops short of localising it there. The classes are assigned by the *solvent* alone, and five of the eight inert-class pairs carry a solute that does have donor-window mass (five primary or secondary amines, 1.2 to 3.8 Å^2 of it against benzene), whose own pure-component segment activity the coefficient scales at infinite dilution. The coefficient therefore has a route into the class its name places outside its remit. For the other three that route is closed: 3-methylpyridine and tripropylamine, twice, carry 0.000 Å^2 there, and the exchange energy is a product of a donor and an acceptor segment, so scaling the coefficient leaves them untouched. Those three are unaccounted, and the improvement on the class as a whole stops short of evidence that the knob acts where it should not. It is silent about a more polar *profile*, a coefficient being a different object from a profile. On the strong donors the error at $c_{\text{hb}} = 0$ is 2.42 against 2.83 at the standard value, so switching the deployed hydrogen-bonding term off lowers the error on the class it governs, on 28 pairs, 13 of them carrying one solvent and 20 carrying two, with no interval, on the set that §3.5.2 declines to stratify at all. Both the two-MSE check and the sweep leave cancellation standing in the regime that generates the drift. They sit on the infinite-dilution axis, they remove the one place it could have been measured, and the cancellation reading remains inferred from the paradox itself. A second reading is live and stays open here. ΔC_p is zero in the deployed ideal term, worth a one-signed median 0.32 in $\ln x_2$ on the test rows carrying a measured melting point (§2.1). The crystal head is supervised toward measured T_m and ΔH_{fus} and is therefore unable to absorb it, so the residual lands in $\ln \gamma_2$. A profile pushed toward its polar wings is what compensating an over-large Φ looks like. The design leaves that entangled with a misspecified activity kernel. What is established is weaker and still consequential: the model's "physical" intermediate departs from the molecule's charge distribution, consistent with cautions that misspecification puts the interpretability of physics-grounded latents at risk,[21,30] and that adapting a representation can itself collapse an otherwise physically-faithful latent.[31] Its *structure* (low-rank rather than diffuse) excludes a random collapse and leaves the structured shrinkage above standing. The full measurement, its nulls and the two controls are in §S5.

What leaves σ under-determined by activity, and what freezing it buys. The freedom that would make such a cancellation possible is the closure's geometry, which we measure in §S5. At infinite dilution the activity map's Fisher spectrum is effectively rank one: participation ratio 1.0 in aggregate and ≈1.4 of 51 per molecule, the resolved direction lying on the hydrogen-bonding wings of the profile. The end-to-end head can therefore move σ along the remaining, low-sensitivity coordinates at little cost to the fit, which is the σ-profile form of Lemma S1. Recovery of the physical profile remains possible in principle for all that geometry shows. The sign of grounding turns on *both* axes, and on which sense of grounding is meant. Freezing the σ head at its supervised warm-up state instead of letting it drift end to end cuts the reference-input penalty by 65% (ΔMAE from +0.40 to +0.14 over n=5571 matched test rows). Those are a separate matched pair of runs, distinct from the +0.41 arms of §3.1: one base, at a shorter P1 than the pinned schedule, resumed twice with the freeze off and on, so a single pair with one run per arm. The 65% is a difference between two runs rather than a mean over any. A near-physical intermediate makes the pipeline far more robust to substituting the reference σ. Supervision by itself stops short of a *flip* in the sign here, the penalty staying positive, the closure still being the low-fidelity COSMO-SAC-2002 and the warm-up σ itself only partly physical.

The nearest published construction, which reports the opposite sign. TeNNet-SAC[8] is this pipeline's closest relative in print: a σ-profile predictor feeding a COSMO-SAC-shaped segment-activity model, with its profile and geometry predictors trained on quantum-chemical σ-profiles and geometries for 39,745 molecules and its segment-activity predictor pretrained on a million synthetic points and then fine-tuned on experimental activity coefficients. That fine-tuning stops short of end to end: it updates the output head of the segment-activity predictor, and the profile and geometry predictors stay frozen. It reports accuracy comparable to COSMO-SAC in its base form and, after that fine-tuning, consistently better than COSMO-SAC across its benchmarks. Three differences separate it from what we measure here, and only the last is a hypothesis. The first is which operation is scored. TeNNet-SAC's gain is a training-time gain from supervising and then fine-tuning a learned profile, and that is the operation this work also finds positive (+0.20, §3.1). The operation we find negative is substituting a tabulated profile for the learned one at inference, which TeNNet-SAC leaves unreported. On that reading the two results are compatible, and the frozen predictors sharpen the point: the one published construction that keeps a σ-profile intermediate and succeeds both supervises that intermediate externally and keeps the activity model's gradient away from it. What we measure here is the complement, an intermediate trained through the closure with neither safeguard, and it stops reporting the quantity it is named for (§2.4). A third design avoids the question by removing the intermediate. HANNA[32] parameterises the excess Gibbs energy directly from SMILES with no quantum chemistry, hard-coding the pure-component limit, the Gibbs–Duhem relation and permutation symmetry in the *architecture* rather than the loss. It reports higher accuracy than UNIFAC over 317,421 mixture points. Between them the two published designs bracket what this work measures: admissibility can be bought with quantum labels or built into the functional form, and this model does neither. The second is that it carries a geometry predictor for molecular volume and surface area, so its composed model can evaluate the combinatorial contribution this one cannot (§2.3). The third is that its segment-activity kernel is trainable where this one is fixed at the 2002 parameterisation. If the sign of the substitution turns on whether the kernel can move, our finding is about a fixed 2002 kernel rather than about routing a prediction through a physical intermediate, and §3.4 already records that no grounding gap has been measured under any modern closure. Measuring it here would take the same two arms retrained against the 2010/dispersion kernel end to end, which is limitation (x)'s named run and remains to be made. Reconciling the two through profile fidelity instead would take a σ-profile departure on TeNNet-SAC's side well below the $36 \pm 1\%$ this pipeline reaches under supervision and the $51 \pm 2\%$ it reaches after fine-tuning, and that departure goes unreported there.

Whether a downstream correction repairs it. If the closure binding were merely an inaccurate point estimate, a learned outputspace correction on top of the physics prediction should recover accuracy. It fails to. An additive bounded residual with a learned confidence gate leaves the gate effectively shut, and forcing it open removes the systematic *bias* ($+0.45 \to +0.06$) while recovering little *accuracy* (R^2 $0.236 \to 0.274$ on the same rows), as expected if the closure rather than the estimate is the binding constraint. Both arms run at seed 42 alone, and this is a statement about the one correction we tested: an additive bounded residual on the output, gated, on this closure and this corpus.

The learned profile is chemically inadmissible in the window the closure's hydrogen-bond term reads. COSMO-SAC's hydrogen-bonding contribution acts on segments beyond $|\sigma| = \sigma_{\mathrm{hb}}$, and the 2002 kernel assigns a segment to the donor side by that threshold alone, with no atom typing. Donor-window area is therefore a statement about surface charge density rather than a hydrogen-bond-donor count, and 264 of the 1003 molecules in the reference tabulation that lack an N–H, O–H or S–H have area there, acetonitrile's 6.71 Å^2 standing above ethanol's 6.20. What carries the argument is the molecule-matched comparison: cyclohexane, acetone, toluene and tetrahydrofuran each carry 0.000 Å^2 of donor-window area in the reference tabulation, exactly zero rather than merely small, against 15.75 for water. The learned profile carries 44.0, 32.4, 24.5 and 50.7 Å^2 in that same window (between 23% and 36% of each molecule's total surface). Every row of the scored set carries mass on both sides of it, so the hydrogen-bond kernel is active on every pair regardless of chemistry. We read those areas off the one trained COSMO-SAC head the tree holds, a retired run trained at 8 segment iterations and scored at 30, and scoring MAE 2.61 at $R^2 = -0.31$ on the scaffold split, worse than every arm of Fig. 2. The null arm that separates architecture from training sits in the same deposit, and we report it here because it exists. Measured as a fraction of profile mass in the same extreme bins, where the reference tabulation's median is 0.000, an *untrained* network of this architecture reaches 0.212 against the trained head's 0.189. The window is full before any training, and training emptied it slightly rather than filling it.

Two causes are separable in principle and stay entangled here, and the head this is measured on decides which is live. Its σ-supervision stream ran for zero steps, so the profile went entirely unsupervised. §S6.2 is the reason that matters: solubility alone underdetermines the profile's shape, so an unsupervised profile is free in exactly the bins that carry no constraint. That accounts for a quantity of this size. The parameterisation accounts for something much smaller and of a different kind. The profile is emitted through a softmax over the 51 bins, which returns strictly positive values where 57% of the reference table's bins, over its 1424 profiles, are exactly zero, so supervision at any strength leaves the window occupied. A softmax reaches 10^{-6} at a logit gap of ten, however, and a floor of that size is four orders below what is measured. The measurement therefore leaves both causes unbounded on their own. What it does establish is that the window is occupied in a trained arm and that the closure's hydrogen-bond term is consequently live on every pair, which is a defect whichever cause carries it. A supervised arm separates the two, and a head able to emit zeros removes the architectural half.

3.4 What the substitution does not decide, and what would settle it

Which of the two channels the substitution fails through stays open here. Drug-like solutes are essentially absent from the VT-2005 set, so the substitution falls almost entirely on the *solvent* channel, and the both-channel substitution the attribution would need covers ≈7 solutes. Enlarging that set would mean computing new reference profiles rather than looking them up.

Why the substitution hurts stays open as well. The reading it invites is that handing the closure a faithful input surfaces a misspecified map rather than repairing it, so that the reference profile is the better input only where the map that consumes it is faithful. That reading is untested here, for a stronger reason than a test that came back empty: on the solubility axis the two terms of the split stay fused, and the solubility axis is where we measured the substitution of §3.1. The quantities the reading is about (a fixed map's own error, and what its inputs fail to carry) therefore lie beyond observation on the rows §3.1–§3.3 report.

§3.5 measures a second axis rather than explaining the first. It uses a different reference database, a different error functional, no trained model at all, and a row set disjoint from the rows above. The two are measured in different regimes on largely disjoint chemistry, and the measurements in this paper leave them unconnected. Until the connecting experiment is run, "the substitution hurts *because* the map is misspecified" remains untested.

Two experiments would settle it, and both remain to be run. The first is the transfer back to the paradox's own regime (finite composition and all γ, where the substitution is measured, rather than infinite dilution, where the split is), which is limitation (i) of §3.7 and the principal open question here. The second is limitation (x), and one part of it sits outside the fidelity contrast of §3.5.3: that contrast compares *oracle-input* activity error across kernels, so $\mathcal{G}$ itself stands measured under no modern closure on either axis, and whether the *solubility* paradox survives a 2010/dsp closure is untested. Both runs need a typed σ-profile head trained end to end through the 2010 kernel, and the solubility one additionally needs a UD $\cap$ BigSolDB matched set, which is unavailable here.

3.5 A second axis: bounding a deployed COSMO-SAC against its inputs

Our estimator is the binning bound, read at the one estimator cell fixed in advance in §2.5 and reserved to this section: eight equal-count bins of the conditioning variable, the residual-only convention, the unbiased within-bin variance, and the admissibility clauses (a)–(c) that decide which strata may be read at all. We fixed all four before computing any margin, and §3.1–§3.4 are measured with other instruments. The decomposition of §2.1.1 separates a fixed closure's error into a part in the map and a part in its inputs, and its output is a map over chemical strata rather than one verdict for a set, for the reason §3.5.1 establishes. It runs on the two data sets of Table 1, which we keep apart throughout.

One reading rule governs every margin below. The comparison is one-sided in both directions it can be read: a positive margin establishes $B_{\text{closure}} > B_{\text{insuff}}$, while a margin at or below zero is a failure to separate the two terms and licenses nothing, leaving their equality open. A margin's size is a *lower bound* on $B_{\text{closure}} - B_{\text{insuff}}$ and not an estimate of it (Lemma 1), so an interval quoted on a margin bounds that bound and not the difference. And B_{closure} throughout is the error of the fixed map *read at a computed profile*, so a positive margin establishes a property of that composite and not of the kernel alone (§3.5.3). The one cell serves both sets, both kernels and every stratum, so every cell below was fixed before the number it returned was seen.

3.5.1 The broad IDAC set (n=477)

The closure reads a temperature as well as a profile pair, so the conditioning variable is its full argument $(z^\star, T)$: the concatenated UD profile pair with T, 103 dimensions against n=477 ($\dim/n \approx 0.22$). The target m is the experimental $\ln \gamma_2^\infty$, on one data set and one profile database, UD throughout. Its chemistry is small—every solute sits at 12 heavy atoms or fewer (median 7), against a median of 14 and a maximum of 122 for the drug-like solids of our own corpus, and 0.46% of corpus rows carry a solute from this set—so *broad* names the reach of the γ^∞ corpus the kernel is calibrated on and not the chemistry a solubility model is deployed against: it covers the solvent side of deployment and leaves the solute side open. It is matched to UD by InChIKey with a first-block fallback, looser than the VT-2005 set's rule, and a displaced $z^\star$ inflates B_{closure} and the binned $B_{\text{insuff}}^{\text{up}}$

together, so its direction on the margin is unsigned; what carries the point is the re-estimate, the margin holding on the 465 exactly matched rows (§S8).

Every MSE, bound and margin on this axis is in squared $\ln \gamma_2^\infty$ units, a different axis from the $\ln x_2$ one the MAEs of §3.1 are measured on. The *2002, all channels* arm scores MSE = 1.902 against $\mathrm{Var}(m) = 4.85$; the binning bound gives $B_{\mathrm{insuff}} \lesssim 0.70$, hence $B_{\mathrm{closure}} \gtrsim 1.21$ and a margin of +0.51 for the set taken whole, positive in every cell of the bin-count × variance × convention grid and with two of its three bootstrap clusterings extending just below zero (§S3.3, §S3.11). The instrument's output is nevertheless the map below rather than that aggregate, because the aggregate is not stable under composition. Dropping the two chemistries the error concentrates in, the glycol-ether solvents and the rows carrying water as the solute, takes +0.51 to −0.12 on the 258 rows that remain; the pair unit alone gives +0.27; the two together give −0.38 on 136 pairs (+0.95, +0.28 and +0.01 under the full convention). One publication carries much of it: `10.1021/acs.jced.7b00114` (37 rows, six pairs, 32% of the set's squared error) takes the margin to +0.18, though the sign survives that deletion and each of the other fourteen (§S3.7). It reverses only under the compounded cut. What we report rests on the binning bound (Table S9); the corroborating estimators say that bound is slack rather than that B_{insuff} is negligible (§S3).

The map, and how much of it survives the rule. We run the instrument over 59 strata of the broad IDAC set (477 rows, 185 pairs, 78 solutes, 36 solvents, 15 source publications). Fifteen have $n \geq 40$ and are boundable, the only ones whose margins the rule can act on; Tables 3 and 4 print all fifteen, and §S3.8 the other 44, the 24 carrying no estimate at all included, each with the test it fails. Six of the fifteen are *admissible*, and clause (c) makes those six three distinct row sets, of which one is both positive and left in place by its residual test. **That row set, the glycol-ether solvents, is the one the admissibility rule leaves standing**, at a margin a chemistry-blind relabelling of the same molecules reaches in 10.1% of draws (below). Its clusters are few and its chemistry narrow, and both belong in the sentence that states it: 182 rows over 43 pairs carrying 45% of the set's squared error, over 4 glycol solvents of one homologous series and 19 solutes, every one a hydrocarbon or a halobenzene, two of three publications carrying 177 of the rows. On them the closure's own error exceeds what input insufficiency can account for by at least +2.04, 90% interval [+1.25, +2.87] at the residual-only convention and +2.94 at the pair unit under the full one, positive in all four unit × convention cells at $P > 0.95$ and under every single-publication deletion, the worst leaving +1.60. Deleting each glycol in turn is the sharper test, three publications mapping onto four molecules: the sign survives all four, while the magnitude runs over a factor of two and is set by the shortest oligomer (§S3.3). That spread, and not the interval, is the honest width on the solvent axis, the bootstrap resampling the same four molecules the deletions delete. The margin survives the loss of the alkane solutes (74 rows, +1.40) and every cell of the bin-count sweep, the analyst choice with the largest leverage here (§S3.9, Table S6).

That sweep varies the binning bound's own settings and not the family of estimator. A pre-declared cross-fitted regressor, tested after this analysis was fixed, returns a uniformly *looser* bound—+1.76 on this stratum, and the chemistry-blind null moved the *wrong* way (§S3.12; note *e* of Table 3)—while under the declaration's conservative source-grouped folds it fails outright, at $R^2 = -0.60$ out of fold and a bound exceeding the stratum's own total error, which B_{insuff} cannot do: there no admissible positive row set survives, and those margins are uninformative rather than reversals. The declaration required that estimator to be adopted UNCONDITIONALLY if it passed a validity gate that reads no margin, and it passed. Every margin printed in this paper is nevertheless the binning bound's. That deviation runs in this paper's favour, and it is a choice made after the two bounds were compared, which is what a declaration exists to prevent. The rule was written against one hazard—taking the smaller bound cell by cell once the cells are seen, which is biased low—and left unanticipated that one valid bound would be uniformly *looser* than the other, where obeying it prints a bound valid and strictly less informative: every claim weakens and none becomes more correct. What limits the deviation is that the number it followed is one global comparison of two bounds on one row set and not a per-cell selection, and that the poorer cross-fitted map is printed cell by cell in §S3.12, where the declared rule can be held against it. The sweep therefore establishes robustness within the binning bound, not across estimators.

The row set clears no multiplicity null, and that belongs beside the finding rather than in a limitation: it is by construction the survivor of a 59-fold search. Our null relabels the chemistry blindly—each solvent and solute keeps its rows and takes another's class labels, and the stratum family is rebuilt and put through the same rule. Over 2000 such relabellings, on all 477 rows and at the cell the map deploys, its six admissible strata are above chance ($p = 0.01$) and its two admissible positive row sets are not ($p = 0.286$), and one row set or more is certified in 58% of draws. The size is the sharper statistic and it is not sharp enough: a certified margin of +2.04 or more arises in 10.1% of all draws, 17% of those that certify anything. What the instrument licenses is the separation on the rows it was computed on; as a chemistry-specific effect it is unestablished against the search that found it. Each p is a frequency and not a bound in either direction, for two biases the null measures rather than argues: provenance survives the permutation, in two halves that run in opposite directions and stand unnetted, so the 10.1% in particular cannot be read as conservative; and the set taken whole has a positive margin at this cell, so a randomly assembled stratum inherits one on average, which inflates the 58% and not the 10.1%. The exactly matched null would permute chemistry and provenance jointly, which these data place out of reach, the two being confounded in them (§S3.6).

What settles a selection objection is the same measurement on rows the search never saw, so we ran one, on the PGL 6th-edition IDAC database—another group's compilation, sixteen times the broad set's rows on a different profile database—scored by the same deployed closure (Table 2). The row set, the estimator cell, the admissibility rule and what would count as confirmation in each direction were fixed before any margin was computed (§S3.7). **The chemistry could not be tested there**, for a reason about the data and not about the number: the cell the finding is about, a glycol-ether solvent with an *organic* solute, is empty there. *Not testable* is the third verdict the pre-declaration lists and counts as non-confirmation, and it follows from the row inventory alone, so what the pre-registration delivers here is the absence of confirming data and not a failed replication. The nearest cell that database carries is water in glycol-ether and polyol solvents, which the rule refuses out of sample for the reason it refuses water-as-solute in sample, so its margin licenses nothing and the sign stays out of risk. Over that database's ten solvent classes, stated as description and not as a test (Table S10), every class either fails the rule or leaves the two terms merged, and every margin above the one admitted positive class sits on a class the rule refuses: out of sample the glycol ethers sit away from where this closure errs hardest, and outside what the rule can read.

That row set is also the whole of the separation this paper reports, and "COSMO-SAC error exceeds input error" would carry it past five boundaries, each of them measured here: the map's other strata, both data sets taken whole, the solubility axis of §3.1 where the two terms stay merged, the kernel as against the composite it is deployed in, and the chemistry-blind null the margin sits at the ninetieth percentile of, on the one chemistry for which out-of-sample rows are missing.

Which chemistry does not separate. Three of the six admissible cells are that same row set at three granularities; of the other two row sets, the alkane solutes are positive and demoted to the glycol ethers by clause (c), and the acceptor-only aprotics pass the rule and leave the terms unseparated, the instrument being one-sided. That is the whole of the uniqueness

Table 3: **The map: every stratum the fixed estimator can bound**: the 15 of the 59 with $n \geq 40$, at the pre-declared cell (eight equal-count bins of $g(z^\star)$, unbiased within-bin variance, row unit, residual-only convention), applied identically to all of them. *full* is the same cell under the full combinatorial convention. Every margin, and the MSE and bound behind it, is in squared $\ln \gamma_2^\infty$. The last column states what the row shows or the test it fails. The other 44 strata, the 24 of them carrying no estimate at all included, are in §S3.8 at this same cell.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solvent class								
glycol ether	182	43	3 (P8, 0.53)	+2.04	$[+1.25, +2.87]$	+2.23	4	the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	$[-0.15, +0.39]^{0.7}$	+0.57	1	the margin changes sign when one publication is deleted: P2 takes $+0.046 \to -0.075$; P14 takes $+0.046 \to -0.194$
aprotic acceptor	57	57	4 (P7, 0.32)	−0.19	$[-0.4, +0.1]$	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
Solvent class, coarse								
H-bond donor and acceptor (O)	328	75	8 (P8, 0.40)	+0.53	$[-0.2, +2.2]$	+1.34	3	the margin changes sign when one publication is deleted: P15 takes $+0.527 \to -0.417$
H-bond acceptor only	70	59	5 (P13, 0.96)	+1.43	$[-0.30, +6.21]^{15}$	+0.77	0	the margin changes sign when one publication is deleted: P13 takes $+1.434 \to -0.192$
Solvent family, fine								
glycol ether	182	43	3 (P8, 0.53)	+2.04	$[+1.25, +2.87]$	+2.23	4	the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water	111	15	3 (P14, 0.75)	+0.05	$[-0.15, +0.39]^{0.7}$	+0.57	1	the margin changes sign when one publication is deleted: P2 takes $+0.046 \to -0.075$; P14 takes $+0.046 \to -0.194$

[a] Number of source publications, then the largest one and its share of that stratum's squared error. The keys: P1 `10.1016/j.fluid.2013.06.028`, P2 `10.1016/j.fluid.2015.01.010`, P3 `10.1016/j.fluid.2015.11.037`, P4 `10.1016/j.fluid.2016.08.030`, P5 `10.1016/j.fluid.2016.10.009`, P6 `10.1016/j.jct.2013.05.006`, P7 `10.1016/j.jct.2015.12.034`, P8 `10.1016/j.jct.2016.10.013`, P9 `10.1016/j.jct.2018.05.003`, P10 `10.1016/j.jct.2019.05.011`, P11 `10.1016/j.tca.2012.02.005`, P12 `10.1016/j.tca.2015.11.010`, P13 `10.1021/acs.jced.7b00114`, P14 `10.1021/acs.jced.9b00726`, P15 `10.1021/je020196d`. [b] Two-way solute $\times$ solvent cluster bootstrap, 5th and 95th percentiles. Row-identical strata are one row set: drawn once and printed once, under the row set's first name above and restated under the rest (10 of 43 row sets, 16 rows). Each prints at the precision its own draw count supports, decided cell by cell over 200 seeds and verified on the run that bought it. 2 carry two decimals (organic solute; glycol ether, the latter at 1.2×10^8 draws) and 9 carry one, both endpoints of each standing four Monte-Carlo standard errors clear of the nearest rounding boundary at that precision. For 11 no precision this table can print is determined: those carry two decimals with a superscript giving their endpoints' Monte-Carlo standard deviation in units of that last digit, and the superscript is the warning rather than a refinement. A further 2 (aldehyde; halogenated | water as solute) no seed moves at all—each has one cluster on one of its two axes, so the resample is one-way over three clusters and about 10 distinct compositions exist: a bootstrap with nothing left to resample, not a converged one. The per-cell verdict, its draw count and its cost at either precision are on the deposited record (§S3.8). [c] In how many of the four unit $\times$ convention cells the stratum is both boundable and sign-stable under deletion of each contributing publication; the rule of §2.5 requires all four, which six of the 59 satisfy, in three distinct row sets. [d] The condition the row is in, rather than a label for it. A one-source cell is untestable, and a deletion leaving fewer than 16 rows leaves the cell undefined, so both read as untested rather than as passes. A margin is one-sided: positive puts $B_{\text{closure}} > B_{\text{insuff}}$ on those rows at that cell, while a margin at or below zero is a failure to separate and licenses nothing. [e] The search behind every row, in the two nulls of the paragraph below. Under the binning bound of record, a chemistry-blind relabelling of the same molecules certifies at least one row set in 58% of 2000 draws, reaches +2.04 or more in 10.1% of all draws and in 17% of the draws that certify anything, and separates the six admissible strata from chance ($p = 0.01$) while leaving the two admissible positive row sets at chance ($p = 0.286$). Under the pre-declared cross-fitted estimator, on the 473 rows it scores, the same relabelling reaches the observed maximum in 13.7% of all draws and in 54% of the certifying ones, against 9.4% and 16% for the binning bound on those same rows.

Table 4: **The map, solute axis**: the rest of the fifteen boundable strata, at the same estimator cell and under the same rule as Table 3, whose notes apply here unchanged. The alkane row is the second admissible positive row set, and clause (c) demotes it to the glycol ethers.

Stratum	n	pairs	sources[a]	margin	90% margin[b]	full	cells[c]	what it shows, or the test it fails[d]
Solute role								
organic solute	440	179	14 (P8, 0.35)	+0.18	$[-0.23, +1.35]$	+0.82	2	the margin changes sign when one publication is deleted: P15 takes $+0.184 \to -0.094$
Solute family								
alkane	129	45	7 (P15, 0.53)	+1.38	$[+0.8, +2.3]$	+1.72	4	the sign holds under every deletion, but 84% of these rows are the glycol ethers' and the 21 outside give -0.22, so this margin does not stand outside the glycol ethers: demoted to them by clause (c); 10% shared with the aprotic acceptors
aromatic hydrocarbon	71	32	7 (P15, 0.53)	−0.19	$[-1.0, +0.7]$	−0.07	0	the margin changes sign when one publication is deleted: P6 takes $-0.188 \to +0.081$
ketone	63	11	3 (P14, 1.00)	−0.53	$[-0.85, +0.63]^{1.8}$	−0.33	0	too few rows remain after deleting P14 (1 of 3 deletions), so the sign is untested there
Solvent class × solute role								
glycol ether | organic solute	182	43	3 (P8, 0.53)	+2.04	$[+1.25, +2.87]$	+2.23	4	the closure's own error exceeds the bound on its inputs, and the sign holds when any one publication is deleted; the same rows as the two other glycol-ether cells, 59% shared with the alkanes
water | organic solute	111	15	3 (P14, 0.75)	+0.05	$[-0.15, +0.39]^{0.7}$	+0.57	1	the margin changes sign when one publication is deleted: P2 takes $+0.046 \to -0.075$; P14 takes $+0.046 \to -0.194$
aprotic acceptor | organic solute	57	57	4 (P7, 0.32)	−0.19	$[-0.4, +0.1]$	−0.15	4	the sign holds under every deletion, but the margin is not positive, so the two terms are not separated here—a bound from above cannot show the inputs are at fault; the same rows as the other aprotic-acceptor cell, 23% shared with the alkanes
The set as a whole								
whole set	477	185	15 (P13, 0.32)	+0.51	$[-0.1, +2.2]$	+0.95	3	passes here and fails at the pair unit (residual-only convention), where deleting P13 takes $+0.270 \to -0.119$

claimed here—the one row set admissible, positive and left standing by clause (c), narrower than being the sole admissible stratum or the sole one sign-stable under every deletion—and which cell each remaining stratum fails on is in the last column of the two tables and in §S3.8, none of them used to explain a reversal.

Why the output is a map. $B^{\mathrm{up}}_{\mathrm{insuff}}$ is nearly the same in every stratum while MSE is not—a factor of 2.0 against 25 over the three classes the design can bound, 6.8 against 51 once the three that carry an estimate are added, that 6.8 resting on a mono-alcohol figure the artifact flags as an invalid bound on a stratum already excluded—and against 2000 random partitions of the 477 rows into blocks of the observed class sizes the chemical partition's range in MSE exceeds every draw while its range in the bound is unremarkable ($p = 0.26$), their ratio giving $p = 0.02$ (§S3.8). That is the estimator showing through: each stratum is summarised by eight quantile bins of one scalar, so the bound measures the residual spread of m inside such a bin, a property of the binning and not of the chemistry, while the error moves independently of it. An aggregate margin is therefore a composition-weighted average of a term free to move 25-fold offset against one that barely moves, and it will report a binding closure for any set whose composition happens to include a chemistry the closure fails on—a statement about the aggregate and not about the bound, which stays valid in each stratum where it is computed.

3.5.2 The VT-2005-matched set (n=60)

This set carries the paper's other reference-matched analyses and is far weaker: all 60 rows sit at 298 K so that $\dim(z^\star)/n \approx 102/60$, methanol occupies 25 of them, and its 17 solvents fall in six of the nine classes, each short of the n=40 a bound needs, so it supports no stratified bound and we state no stratum of it as a finding. Its aggregate margin of +0.05

collapses—under deletion of the two leverage pairs, under dropping methanol, under blocked out-of-fold estimators and under 11 of the 60 single-pair deletions, where on the broad set every single-pair deletion leaves the sign intact—and what a margin here would bound is a label systematic tracking $z^{\star}$, which the unbiased within-bin variance leaves *open* (§S3.11). Of the 60 pairs, 57 also appear in the broad IDAC set at 298 K under UD profiles, so the two are overlapping samples rather than independent ones. This is the sub-design VT-2005, and therefore the paradox's own oracle, is defined on, and the agreement between the two sets' error orderings is a consistency check rather than corroboration.

3.5.3 Whether a better closure moves the ceiling, and where this one fails

If the ceiling is the closure, the direct remedy is a better closure. The standard 2002 → 2010 step delivers one on the VT-2005-matched set, more than halving the error under the full convention ($P > 0.95$), and withholds it on the broad IDAC set, where it scores −0.99 at $P \approx 0.2$ under the residual-only convention and leaves both an improvement and an equality unestablished in each of its three cells (Table S8). A block split places that reversal in the *chemistry* and not in the profile database or the temperature range: the *2010, dispersion off* arm wins on the 57 pairs shared with the smaller set (1.78 → 0.77, $P > 0.95$, full convention) and loses that win on the complementary 420 and again on the 80 of those at 298 K, the block the temperature half of the claim rests on and the weakest of the three. Those 57 pairs are broad-set rows too, so they are an overlapping sample and not an independent one, and that split is the whole of the evidence for the reversal being chemical. We claim only that **the fidelity lever we can demonstrate is confined to the smaller set's chemistry**, and that null is additionally a small-molecule and glycol-heavy one, so on drug-like solids the lever is untested rather than absent. Under the 2010 kernel's own typed latent, 306-dimensional and so several times worse resolved, the floor checks return +1.29 on the broad set against −0.77 at $P \approx 0.3$ on the smaller one, of which the first stands: loss of resolution rather than a demonstrated flip to the inputs, which would need a *lower* bound on B_{insuff} (§S6.6).

Where B_{closure} dominates, on the one row set the rule leaves standing, the ceiling is in the map *as it is deployed*: the fixed kernel together with the one computed conformer it reads, which is the composite B_{closure} is the error of and the object the pipeline runs on. Which of the two carries the error this section leaves open, and dominance leaves open too whether the composite is *fixably* wrong—a fitted kernel residual does not transfer across solvents on the one fold design free of leakage, and that result is a null and not a bound, so B_{closure} is an error a handful of kernel parameters can absorb *within* a solvent and whose correctability across solvents we leave unshown. Where it fails is chemically placed: on the VT-2005-matched set 90% of the squared error falls on strong hydrogen-bond-donor solvents and the closure is near-exact on acceptor-only ones, the chemistry the documented single-parameter H-bond term governs and the term the donor/acceptor-typed 2010 closure was built to replace.[27,33] That the 2002 kernel fails there is established and the localisation restates it; what is new is that on one such chemistry the failure is bounded against what the closure's inputs leave unresolved. Half of that reading carries to the published evaluation of Table 2 and half stops there, and on the broad set neither it nor its replacement reproduces, so we state neither. It places the error in a *chemistry* and not in one component of the composite, those solvents' reference profiles being single gas-phase conformers too; of that, the conformational part is bounded rather than open, since replacing the four glycol-ether references with Boltzmann ensembles moves the stratum's margin from +2.04 to +2.27, and every substitution reported here stays scoped to the VT-2005 tabulation's level of theory (§S3.10).

That COSMO-SAC-2002 is a low-fidelity closure is the contribution rather than a qualification of the result: it is what a practitioner adopts by default, a closure's fidelity is *unknowable in advance* on a new system, and the field routinely composes an expressive encoder with a fixed mechanistic map and assumes the map is good enough. Where that assumption fails the learned latent silently departs from the quantity it is named after (§3.3), and an external oracle is what makes the departure visible. Outside that one row set the map reports cells it is unable to resolve, which is weaker than cells in which the closure is sound. How large an exposure the choice of profile database is, and why that arithmetic sizes it without bounding its share, is in §S6.6.

3.6 The same split on a second closure: pKa through a fixed Hammett relation

The instrument generalises beyond COSMO-SAC, and a second domain runs it where the sign of grounding can be predicted before the measurement. The closure is a Hammett linear free-energy relationship $g(z^\star) = \mathrm{p}K_{a,0} - \rho\,\sigma^\star_{\mathrm{H}}$ in a substituent constant, with $\mathrm{p}K_{a,0}$ and ρ the tabulated Hansch–Leo–Taft reaction-series constants, one literature pair per scaffold: $(4.20, 1.00)$ benzoic acid, $(9.99, 2.23)$ phenol, $(4.60, 2.77)$ anilinium and $(5.25, 5.90)$ pyridinium. Those pairs come from the literature and stay unfitted on these data, so g is fixed in the sense Lemma 1 requires, and the same g runs inside every split. It reads two things, so $z^\star$ is the pair $(s, \sigma^\star_{\mathrm{H}})$: the series s, which selects that literature pair, and the substituent constant at the tabulated value the rule below assigns. The constant is the one learned quantity, the learned intermediate $\hat{\sigma}_{\mathrm{H}}$ being that constant predicted from structure. The external reference is the Hansch–Leo table,[34] printed in full in §S7, and the data are the OPERA experimental compilation[35] under a uniform, closure-independent quality filter that drops pre-ionized microstates.

We assign the reference constant mechanically, by a rule that reads structure alone and never the residual (§S7), and that rule is also what defines the two poles. A record enters under one of the four reaction series (benzoic acid, phenol, anilinium, pyridinium), which fixes its $(\mathrm{p}K_{a,0}, \rho)$. Then $\sigma^\star_{\mathrm{H}}$ is the sum, over the substituted ring positions, of the Hansch–Leo constant of that substituent *at that position*, σ_m meta and σ_p para, taken from a fixed list of nineteen. An ortho position carries no tabulated constant and enters the sum as zero, and so does a substituent outside the list. The first pole (the *clean* one below) is the 158 records whose every substituted position is meta or para with a tabulated substituent. The *degraded* pole is the 846 where at least one position is neither (571 of them carrying an ortho substituent, the other 275 an untabulated one), and there $\sigma^\star_{\mathrm{H}}$ is the partial sum over the positions it can see. The split cuts across benzenoid and heteroaromatic alike: all four series occur on both sides, 32 of the clean pole's records being pyridinium.

The two classic failure modes of that relation fall onto the two channels of Lemma 1. *Resonance saturation*, a curvature beyond the reach of a linear closure, lands in B_{closure}, while what the zeroed positions carry (ortho field and steric effects, and whatever an untabulated substituent contributes) is information the intermediate lacks and lands in B_{insuff}. The two poles differ accordingly. On the clean one, the chemistry the relation was built for (n=158), the closure predicts to RMSE 0.55, and the two available bounds on B_{closure} disagree about what that leaves. The binning-free floor MSE $- B^{\mathrm{up}}_{\mathrm{insuff}}$ is -0.2 (95% CI $[-0.4, -0.0]$), and it establishes nothing about this closure. The out-of-fold $B^{\mathrm{up}}_{\mathrm{insuff}}$ is 0.500 against an MSE of 0.302, so that difference is negative whatever the closure. What those 158 rows across four series show is that a flexible regressor on the closure's own argument predicts worse than the closure does. The Lemma 1(b) bound is informative and runs the other way. The per-scaffold systematic offsets reach 0.429 in $\mathrm{p}K_a$ on phenol (+0.303 anilinium, +0.034 benzoic acid, -0.198 pyridinium), and the n-weighted mean of their squares bounds B_{closure} below at 0.083, 27% of the 0.303 total, and at 0.184 on phenol alone, half of that series' own 0.371. At least a quarter of the clean pole's error is closure error, so the pole's specification stays in doubt. The quality filter moves the floor and leaves that conclusion. The rule is closure-independent (net formal charge $\neq 0$, applied uniformly to both poles and blind to any residual), and it drops five preionized microstates. On the 163 rows before it the same floor is $+0.55$ rather than -0.2, positive and establishing $B_{\text{closure}} > 0$ by the other route as well (§S7). On the degraded pole (n=846) it degrades to RMSE 2.26, of which the out-of-fold within-series regressor places at most 4.5 of the 5.1 total in input insufficiency and hence at least 0.6 in the closure, both figures one-sided, in the same direction as the solubility bounds. The filter leaves that pole where it was and moves the other by a factor of eleven in MSE, 3.33 to 0.30. Before it the two poles differ by half an order of magnitude and after it by more than one, so a large part of the contrast between them is what the filter makes. The reversal itself is a sign measured within each pole, so it stands clear of that, while the poles' separation in level rests on it, as does the sign of the clean pole's floor.

That estimator differs from the eight-bin estimator cell of §2.5, and the functional differs as well. The two figures just given are floors MSE $-$ $B^{\text{up}}_{\text{insuff}}$ on B_{closure}, which when positive establish $B_{\text{closure}} > 0$. They are not the separation margin MSE $-$ $2B^{\text{up}}_{\text{insuff}}$ of Table 1, which when positive would establish the stronger $B_{\text{closure}} > B_{\text{insuff}}$. Both coordinates of $z^\star = (s, \sigma^\star_{\text{H}})$ are coarse enough to use directly, s taking four values and $\sigma^\star_{\text{H}}$ being a scalar. We therefore estimate the conditional variance as the *out-of-fold* residual mean square of a random forest of pK_a on $\sigma^\star_{\text{H}}$ fitted *within* each series, five-fold and pooled over the series by n. That quantity exceeds B_{insuff} by the regressor's own approximation error, so it leaves a lower figure on B_{closure}. Conditioning on s is forced rather than free, and the conservative label belongs elsewhere. Pool the four series and the same estimator returns 8.3 against the degraded pole's MSE of 5.1. Under that conditioning g is unmeasurable, so Lemma 1 withholds its licence, and the closure floor it leaves would in any case be vacuous. The floor of 0.6 therefore rests on s belonging in $z^\star$, which is what the lemma demands of a closure that reads it. Those folds are free of the leakage §3.5.1 refuses a fitted estimator for on solubility, where a held-out row borrows strength from a near-twin through the representation. Here the regressor reads $\sigma^\star_{\text{H}}$ within a series and nothing besides, so a fold can hand it only what the closure's own argument already carries, and the identity asks of the fold design that a row be held out of the fit that predicts it and no more.

The decisive test is the same oracle swap, measured *in-distribution* across K=20 within-scaffold splits: the learned $\hat{\sigma}_{\text{H}}$ replaced by the tabulated $\sigma^\star_{\text{H}}$ in the same closure (the series label is read from the structure in both arms, so the swap moves one coordinate of $z^\star$), the sign of the change recorded per stratum. It flips. On the clean pole (n=158) the reference *improves* accuracy (MAE 0.48 learned against 0.315 with the reference, a gap of -0.165 at a 90% CI of $[-0.28, -0.04]$, better in 19 of 20 splits and in three of the four reaction series), so the ceiling there is the noisy learned estimate and not the closure, which is an internal positive control on real data. It is a control on the swap's sign and not on the closure's specification: the Jensen bound puts at least a quarter of that pole's error in the closure, so the reference wins there against a closure carrying error of its own. On the degraded pole (n=846) the same swap *degrades* it (0.83 against 1.62, worse in 20 of 20 splits and in all four series, a gap of $+0.79$ at a 90% CI of $[+0.57, +0.95]$), the freely learned constant absorbing what the zeroed positions carry and the tabulated one lacks. The sign of grounding therefore reverses inside one closure, in the direction the two poles' construction fixes in advance, which ring positions the Hansch–Leo table can see, complete on one pole and a partial sum on the other. Remark 1 supplies the vocabulary and not the prediction. That the reversal runs through the insufficiency channel rather than through the closure-error cancellation of §3.3 is unsupported on the clean side, where the Jensen bound above leaves at least 27% of the error in the closure. On the degraded side the attribution rests on the flip itself (the freely learned constant beats the tabulated one there, 0.83 against 1.62, in 20 of 20 splits) and on how that pole is built, $\sigma^\star_{\mathrm{H}}$ being a partial sum over the positions the table cannot see. What the Jensen bound adds there is compatibility with that reading: it leaves at most 87% of the error in insufficiency, and an upper bound is as consistent with none of it as with all of it. The out-of-fold regressor two paragraphs above is a different estimator answering the same question on the same pole, at 4.5 of the 5.1, and it is an upper bound too, so the two agree while leaving the attribution to rest elsewhere. Both signs are one comparison: the sign of the grounding gap is by construction that of the learned arm's error against the fidelity-set oracle error, and what a competence sweep adds is that the oracle side of it holds fixed across the training budget while the learned side moves. The learned arm stays above the clean pole's threshold (≈ 0.315), so grounding helps at all four budgets swept, while the degraded pole's threshold (1.62) is high enough that the learned arm beats it across the whole interpolation range, down to 12% of the data, and grounding reverts to helping only under scaffold *extrapolation*, where the learned constant breaks (2.6±2.0) back above the threshold, in the mean over the six splits, three of which still have grounding hurting. **The flip is an in-distribution result and the scaffold protocol does not certify it.** Re-running the same K=20 certification on a scaffold split rather than a stratified one, the reference helps on *both* poles: the clean pole's gap is -0.61 and the degraded pole's -1.10, with a series-cluster interval of $[-1.86, +0.37]$ and $P = 0.28$ there, so the degraded pole's "grounding hurts" half fails to certify, and so does the reversal. We measure every solubility result in this paper on a scaffold split, and this is the one place the protocol changes. What the scaffold run confirms is the mechanism rather than the flip, the learned constant breaking under extrapolation and rising above the oracle threshold on both poles. The clustering unit is the Hammett reaction series, of which only G=4 exist, so the bootstrap resolves series-level sign agreement and nothing finer, read as a sign test on four units: p=1/16 on the degraded pole, where grounding hurts and all four series agree, and $p \approx 0.31$ on the clean one, where three of them do. That test carries the degraded pole and not the positive control, which rests on its splits and its gap instead. The construction, the estimator validation and the measured fidelity sweep are in §S7.

3.7 Scope, accuracy, and limitations

Two measurements, in different places. On solubility, supervising a learned physical latent on reference values during training moves the error without an established sign, while *supplying* those values at inference hurts at every seed; on infinite-dilution activity data, and there on one class of solvent, the fixed map's own error exceeds the bound on what its inputs leave undetermined. The two sit in different regimes on largely disjoint chemistry, so "the substitution hurts *because* the map is misspecified" remains untested (limitation (i)). The finding is one-sided: one row set of the map on which grounding effort is *not* the remedy.

The unconstrained control's representation. If the physics arm's intermediate is a surrogate, the thermodynamics might still be present implicitly in the control, which is free to represent it. It is absent. Our ridge probe of the control's pair representation recovers the model's own prediction at $R^2 = 0.98$ and leaves the measured ideal-solubility term unrecovered. We score it on 13 held-out solutes / 527 rows at zero solute overlap, and that term is 92.2% between-solute variance, so the effective n is 13 clusters rather than 527 rows. The three seeds give $R^2 = -0.506/+0.111/+0.027$ at 42/43/44, with permutation $p = 0.894/0.055/0.177$ against a 1000-draw null. Their between-seed range of 0.616 (sd 0.334) exceeds any one of the three values, so a single seed would have been unreadable and seed 43 alone would have read as a near-miss. The negative therefore falls short of an exclusion, and we leave uncomputed the level below which it excludes nothing. What the deposit does fix is the null's shape: its

95th percentile is 0.105/0.117/0.096 over 1000 draws and its maximum 0.316/0.380/0.346, and each of the three measured values sits at or below its own null's ceiling (which is what the three permutation p above say in another currency). The design therefore excludes nothing below that ceiling, and we ran no power calculation to say how far above it the design reaches. We draw no comparison between the between-seed spread and the null's support: the null is two-sided and reaches below the lowest measured value, so its support is the wider of the two. That is limitation (ix), and two things fix its size rather than the encoder's. The 13 solutes are capped where they are by the reference scarcity of §2.2, and the temperature probe rests on the same 13. And the crystal-slope test is capped by the corpus rather than by the probe: this corpus's empirical $\ln x_2$-against-$1/T$ slope carries $-\Delta H_{\mathrm{fus}}/R$ at $r = +0.24$, so the quantity the probe is asked to find is itself weakly present in the labels it is scored against. What the representation is organised by instead is scaffold identity, whose share of the representation's variance sits at the solute-identity ceiling. That holds on a split where the two partitions nearly coincide by construction, 11 of the 124 test scaffolds holding more than one solute. They agree on 91% of the data, so a share at that ceiling is weak evidence of within-scaffold indistinguishability rather than a measurement of it. A test molecule on an unseen scaffold is a point off the axis the representation is organised along rather than a small extrapolation. That is a description of this representation rather than a diagnosis of the encoder, which the linear probe of §S6.1 leaves unsupplied. The full tables, the controls, the direct within-scaffold measure on the solutes that do share one, and the temperature probe are in §S9.

Where the accuracy stands, and the confound in it. Table 5 places every arm of this work beside the external models, in two blocks that may not be read across. The physics arms trail their own control in the seed mean, the COSMO-SAC arms at each of the three seeds the two families share and the NRTL arm at two of them (Table S3). The two sides are separately tuned and we ran no hyperparameter-matched control, so the ΔMAE of +0.23 between them on the 5608 labelled test rows, and the chemical strata that resolve the same difference along chemical axes (§S6.4, §S6.5), are silent about the thermodynamic bottleneck. We claim no accuracy result in either direction, and the matched control that would repair this is unrun. What the confound leaves untouched is every *within-checkpoint* contrast, where the two sides share one trained model and one tuning (§3.1, §3.2, §3.3, §3.5); the supervision gain, the co-adaptation control and §3.2's shrinkage slopes fall outside it, comparing across trainings or across the separately tuned control. Grounding the crystal factor on external labels is underpowered rather than null: 97.76% of that pool was melting points the model had already been trained on, and where the pool carried something new the crystal head moved, while on the held-out solutes it is the worse of the two and the downstream solubility gap crosses zero once clustered by solute (§S6.3). Temperature extrapolation is the one place the physical *structure* helps, and it helps as a property of the hypothesis class rather than of any arm trained here: fitting below $T \leq 309.75$ K and scoring at $T \geq 330$ K on the same 1751 solute–solvent pairs, so that the split withholds temperature and keeps structure, a structured van't Hoff class reaches an $\ln x_2$ MAE of 0.37 against 1.29 for a random forest over Morgan fingerprints and temperature, each one fit on one temperature cut with no replicate behind it (§S6.1).

The external block is on an easier by-solute split, and the harness for a like-for-like comparison is deposited and unrun. What remains is a scale rather than a placement: this work's control reaches $\log_{10} S$ RMSE 1.00 on held-out scaffolds and its physics arms 1.39 to 1.60, against 0.83 to 0.99 for the two leading direct models and 1.43 to 2.16 for the physics-informed SolProp cycle.[2] All six are as published by Attia et al.,[1] in $\log_{10} S$ RMSE in $\mathrm{mol\,L^{-1}}$, on two other corpora: the SolProp test set, whose solutes are disjoint from its training corpus, and the near-ambient Leeds set of Boobier et al.[36] Sharing no row with these, they set a scale and withhold an ordering; each range is two point values on two corpora rather than a spread, and none of the six carries a published uncertainty. The clip sweep, the random-pair rows and the conversion behind the $\log_{10} S$ column are in §S4.

The sign rule's own prediction. Remark 1 places these negatives in the mature-data, low-noise, well-covered regime, where a prior is predicted to hurt. The synthetic sweep (Fig. S9) is true by construction there, so it verifies the instrument rather than the hypothesis, while the pKa closure exercises the sufficiency channel on real data.

Limitations. All fourteen, keeping the numerals used throughout; §S6.7 carries the qualifying detail for the nine it holds.

(i) The phenomenon and its causal attribution lie in *different regimes*: the paradox where the model operates (finite composition, all γ, n=5608), the attribution and the latent-departure result on infinite-dilution activity data alone (n=477 across temperatures, and the 298 K, low-γ, VT-2005-matched n=60 and n=44 sets). Transfer back to the paradox's own regime is a conjecture and the principal open question here. The n=60 set is all gas chromatography and all from three publications, 33 pairs from one, so a method-stratified or source-clustered re-estimate lies beyond it, and its margin inverts under a broader ThermoML pull where the broad set holds steady. Both sets sit outside the deployment regime (§S8).

(ii) What the convention rule costs the aggregate (§2.5).

(iii) That other cells of the map keep their sign as well (§3.5.1).

(iv) B_{insuff} is upper-bounded rather than point-identified, the design holding no replicates at fixed $z^{\star}$. B_{closure} is the error of a composite, the fixed map together with the calculation that produced $z^{\star}$, and the measurements here leave it undivided between the two and its share unbounded (§3.5.3).

(v) The constant-offset bound's convention-specificity (§2.5).

(vi) The stratification reads structure alone, so every cell was fixed before its outcome was seen. Cells from different axes share rows and are therefore dependent evidence, counted once; granularity is an analyst choice, which is why we report three solvent levels and two solute levels. The per-cell bootstrap frequencies carry no multiplicity control, so one cell's P describes that cell alone, and the admissibility rule is a stability filter rather than a correction for the search. Neither test of the search establishes a chemical effect: under the chemistry-blind relabelling null the shape of the finding actually claimed sits at chance (§3.5.1), and the pre-declared out-of-sample test returned *not testable*. We therefore state the finding at the level the instrument licenses on its own rows and *not* as a multiplicity-controlled chemical effect, and the same holds a fortiori for every cell the map reports and leaves unstated. With B_{insuff} bounded from above alone, every cell is also silent on whether better inputs would help there, so the map's one action is to withhold a chemistry from an allocation of grounding effort.

(vii) The accuracy ceiling standing a factor of five above the label-noise floor (§2.2).

(viii) What the three-seed latent-departure result establishes and where it stops (§3.3).

(ix) What bounds the black-box probe (the probe paragraph above).

(x) What the closure-fidelity contrast leaves unmeasured, and what would settle it (§3.4).

(xi) No accuracy difference between the arms is attributable to the bottleneck (above). Whether grounding helps is shaped by *two* axes, closure fidelity and how the latent is trained, and this work leaves that plane uncharacterised on real systems. The chemical strata form a pattern in the direction the sign rule would

Table 5: Solubility accuracy in two blocks that may not be read across: this work on the solute-scaffold split, external models re-run here on an easier by-solute split. Two sample sizes appear where a row's two metric columns are scored on different rows, one where they coincide. The grounded, ungrounded and σ-oracle arms of the first block come from the five-seed leak-free re-run.

Model	n ($\ln x_2/\log_{10} S$)	MAE $\ln x_2$	RMSE $\log_{10} S$	R^2 $\ln x_2$
This work, solute-scaffold split; mean ± s.d. over seeds: five for the two COSMO-SAC rows, three for the others				
DirectGNN, same encoder, no closure *(control)*	5608 / 5440	1.70±0.03	1.00±0.02	0.42±0.03
TGNN, NRTL closure (physics)	5608 / 5440	1.79±0.07	1.54±0.07	0.34±0.03
TGNN, COSMO-SAC closure, supervised $\hat{\sigma}$ (physics)	5608 / 5440	1.93±0.09	1.44±0.11	0.29±0.07
TGNN, COSMO-SAC, reference $\sigma^\star$ substituted at evaluation	5608 / 5440	2.34±0.11	1.64±0.09	−0.07±0.08
Scale reference on the same solute-scaffold rows				
predict the test-set mean of $\ln x_2$	5608 / 5440	2.32	1.33	0.00
External models re-run on this corpus on a by-solute split, one run each—not a ranking against the block above				
FastSolv, retrained on this corpus, re-scored	10287	1.44	0.84	0.55
SolProp cycle, released model + 3-parameter recalibration	10287	2.34	1.34	−0.18
SolProp cycle, released model, zero-shot	10287	2.36	1.87	−0.47
SolProp encoder retrained on $\ln x_2$, cycle discarded	10570 / 10287	1.62	1.07	0.39

give, but separately tuned pipelines can produce one and a multiplicity control is missing, so §S6.4 records it as the experiment a matched control should run and the diagnostic is *proposed* rather than validated.

(xii) The scaffold-leak guard is a property of the released stream builder rather than a certified property of the runs §3.1 reports, so the 0.20 three-seed supervision gain is the one reported number a leak could inflate. Deleting the eight reachable solutes leaves +0.17 [+0.03, +0.31], which bounds where a leak can act while leaving the build uncertified, as do the arguments that bound the other contrasts (§2.2). The re-run that is certified leak-free returns that gain without a sign.

(xiii) What the deposit leaves uncheckable (the data-availability statement).

(xiv) Two terms the thermodynamics asks for are set to zero in the deployed model, and both are absorbed at training time by the same learned σ head: ΔC_p in the ideal term, one-signed and worth a median 0.32 in $\ln x_2$ where a melting point is measured (§2.1), and the Staverman–Guggenheim combinatorial term, off in every arm the paradox is reported on. Both cancel in the evaluation-time substitution and both survive the training that produced the drift of §3.3, so the mechanism there reaches beyond the activity kernel.

4 Conclusions

Grounding a learned physical intermediate in reference values is two operations rather than one, and here only one of them carried a sign. The operation that failed is the one a practitioner reaches for, because it is the one that runs without retraining. The intermediate paid for the arrangement. Trained through a model beyond its power to correct, and measured on runs of its own, it stopped reporting the physical quantity it is named after wherever a reference existed to hold it against. That is the price of reading a grounded latent as a measurement.

The general statement is a design rule with a test attached. Wherever a predictor composes a fixed physical model over a learned intermediate for which an external reference exists, that reference measures whether the fixed model itself limits accuracy, and where it answers, the answer decides the remedy: a more faithful physical model, the calculation that supplies its reference values included. The answer comes chemistry by chemistry rather than once for a field, and it comes one-sided, so what the measurement returns is the chemistry on which better inputs are not the remedy. What this work establishes about the instrument is the map itself: which cells it can answer in, and at what size. Whether a latent trained through an imperfect model becomes, in general, a surrogate for that model's error is the prediction this work leaves testable.

Author contributions

N. L. Polomoshnov: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. A. V. Rudik: Writing – review & editing.

Conflicts of interest

There are no conflicts to declare.

Data and Software Availability

The analysis code, and those intermediate artifacts small enough to version, are available at `https://github.com/doctawho42/tgnn-solv` (commit **[hash to be supplied at submission]**) under the MIT licence, that commit being the one the reported numbers were produced from; the model checkpoints, the processed split, the full per-arm prediction files and the larger analysis artifacts will be deposited in a Zenodo archive, **[whose DOI is registered at submission]**. The data sources are public: BigSolDB 2.0[24] for solubility, the VT-2005 database for the reference σ-profiles, Ref.[33] as redistributed with Ref.[26] for the UD profiles, ThermoML for the infinite-dilution activity coefficients, the PGL 6th-edition IDAC database for Table 2, and the OPERA compilation[35] for the pKa data. Three disclosed defects sit on the deposited record, in §S8 and §S3: the three-seed surrogate run's per-run split provenance was not logged, one σ-stream build on disk was written against the wrong split files, and the two VT-2005-matched audit artifacts disagree on one out-of-fold estimate. Two further defects are repaired rather than disclosed, each described where its numbers are printed. We re-scored the two FastSolv retraining bundles (Tables 5 and S17, with the diagnosis, the withdrawn values and the control that reproduces them in §S4), and the withdrawn bundle is kept beside the re-scored one. And the truncated seed-44 σ-oracle per-row file has been recovered complete and now stands in the deposit (Table S25, §S8). What that deposit does and does not make checkable is limitation (xiii). Several of the numbers the verdict rests on are checkable by re-running the released code and by that route alone, and its input artifacts are larger than the repository carries. We therefore deposit the two decomposition sets as row-level tables (Data Sets S1 and S2), so those sets and their margins can be recomputed with no code at all. That leaves the provenance of the closure evaluations inside them, and of the trained arms, undelivered.

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Supporting Information Available

The Supporting Information is a separate document. Its sections, tables, figures and equations are numbered S1, S2, ..., and every reference above that carries an S is to it. It contains supplementary methods; the proofs of Lemma 1 and Remark 1 and three further results; per-arm tables and decomposition robustness; the map's 59 strata in full, including the 44 the estimator cannot bound. It also contains further external ablations; supplementary mechanism results; supplementary diagnostics and broad negatives; the second-domain pKa construction; data provenance and reproducibility; the black-box probe tables; the synthetic closure-fidelity family; and the positioning against adjacent literatures. We deposit two machine-readable tables with it: the 477-row broad IDAC set (Data Set S1) and the 60-row VT-2005-matched set (Data Set S2), and their columns are listed in §S8.

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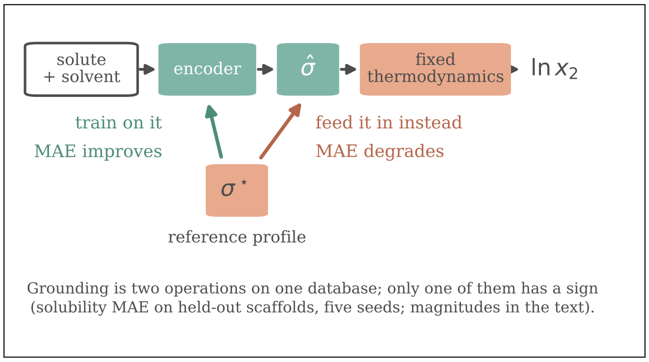
TOC Graphic


solute
+ solvent
encoder
$\hat{\sigma}$
fixed
thermodynamics
$\ln x_2$
train on it
MAE improves
feed it in instead
MAE degrades
$\sigma^\star$
reference profile
Grounding is two operations on one database; only one of them has a sign
(solubility MAE on held-out scaffolds, five seeds; magnitudes in the text).